Kriging Variance Is Not Reconstruction Error: Leave-One-Out-Calibrated Look-Ahead Acquisition for Autonomous Experiments

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Abstract

Kriging-guided autonomous experiments at synchrotron beamlines choose the next measurement where the Kriging variance is largest (SMART) or where it is amplified by the squared surrogate gradient (gSMART). For a stationary covariance this variance depends on where measurements were taken but not on what they returned, so variance-driven sampling is a space-filling design in disguise. We introduce CALM, a calibrated absolute-error look-ahead. Standardised leave-one-out residuals of the unchanged Kriging surrogate are treated as log-χ^2 observations of a spatial variance-calibration field, which a second Gaussian process fits by marginal likelihood. The next measurement is the one that maximises the expected reduction of the domain-integrated absolute reconstruction error under the calibrated model. A scaling analysis shows that this greedy rule targets the error-optimal sampling density, whereas gradient weighting has no equilibrium at discontinuities and pointwise cross-validation criteria over-concentrate. On the five synthetic benchmarks of the original SMART study we use 10 paired seeds and the same `gpCAM` surrogate and optimiser for every method. CALM reaches SMART's final-budget integrated absolute error with 47 instead of 1000 measurements on the 3D Stokes–Einstein diffusion map, and with 228–247 instead of 500 on three maps with discontinuities. Against gSMART with a weight tuned in hindsight, the saving on those maps shrinks to 1.81–1.93×. On these four maps CALM also yields the lowest RMSE and, once its calibration field is used for the predictive variance, the lowest CRPS. It gives no sample-efficiency benefit on a statistically homogeneous smooth map, which a heterogeneity index computed from the learned calibration field predicts in advance.

1 Introduction

Autonomous experiments replace the human operator of an instrument by a decision rule that chooses where to measure next from the data collected so far [Stach et al., 2021, Kusne et al., 2020, Ziatdinov et al., 2022]. At synchrotron and neutron facilities the dominant use case is *mapping*: a focused beam probes a sample or a materials-parameter space at positions $\mathbf{p}$, each measurement is reduced to a scalar $\rho(\mathbf{p})$, and the goal is a faithful reconstruction of the whole response surface rather than the location of its optimum [Noack et al., 2021, Yager et al., 2023]. Because every measurement costs beam time, the figure of merit is the reconstruction error reached for a given number of measurements.

The reference approach, SMART [Noack et al., 2019], fits an ordinary-Kriging surrogate and measures next where the Kriging variance is largest; its gradient-weighted variant gSMART multiplies the variance by $1 + c|\nabla m|^2$ to favour sharp features. SMART proceeds coarse-to-fine and beats naive raster scans in simple settings, and it underlies the `gpCAM` software used at several facilities

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[Noack et al., 2020a,b, 2021]. Its decision rule, however, rests on a quantity that is *not* the reconstruction error. For a stationary covariance the Kriging variance depends on where measurements were taken but not on what they returned. A map with a sharp phase boundary and a map of a featureless slope therefore receive the same design, up to a single global range parameter. Gradient weighting injects data dependence, but through a heuristic whose weight has no natural scale. We show that the weight helps only inside a narrow window that cannot be located without ground truth: smaller weights do nothing, and larger ones trigger, at discontinuities, a runaway in which every new measurement makes the boundary more attractive.

This paper asks how much a Kriging-based autonomous experiment can gain by deciding with an estimate of *where its surrogate is actually wrong*, while keeping the experimenter's surrogate untouched. Our answer is a new acquisition criterion, the *calibrated absolute-error look-ahead* (CALM), and an analysis of when it pays off. The construction has two parts. First, standardised leave-one-out (LOO) residuals of the Kriging surrogate are, under a locally scaled error model, log-χ^2_1 observations of a spatial *calibration field* $\lambda(\mathbf{x})$ that rescales the stationary variance. A second GP fitted to them by marginal likelihood yields a calibrated error model that collapses back to the stationary one when the data show no heterogeneity. Second, CALM chooses the measurement that maximises the expected reduction of the domain-integrated absolute error under this calibrated model, the myopic Bayes-optimal action for the metric used to judge reconstructions. A scaling analysis shows that this greedy rule targets the error-optimal sampling density $\varrho^\star \propto \tau^{2d/(2d+1)}$ in terms of the local scale τ. By contrast, SMART is uniform, pointwise LOO-adjusted variances over-concentrate ($\varrho \propto \tau^{2d}$), and gSMART has no equilibrium at discontinuities.

We evaluate CALM on the five synthetic benchmarks used to validate SMART: two piecewise-sinusoidal maps with step discontinuities, a smooth oscillatory map, a non-differentiable piecewise-linear map, and the 3D Stokes–Einstein diffusion coefficient of nanoparticles in water–glycerol. We use 10 paired seeds, budgets up to 500 (2D) and 1000 (3D), and the same `gpCAM` surrogate and optimiser for every method. Our contributions are:

- **A calibrated error model for a fixed Kriging surrogate** (§2.2): a likelihood-based recalibration of the Kriging variance from closed-form LOO residuals, with known log-χ^2 noise and a single marginal-likelihood selection rule.
- **The CALM acquisition** (§2.3): an integrated, L^1-aligned look-ahead under that model, implemented as a `gpCAM` acquisition callable and optimised with `gpCAM`'s own optimiser, so that it differs from SMART only in the criterion.
- **An allocation theory** (§2.4) that predicts, before any experiment, on which response surfaces calibration helps, and explains the failure modes of gradient weighting and of pointwise cross-validation criteria.
- **A paired, fully logged evaluation** with reproduction of SMART/gSMART on `gpCAM`, modern predictive metrics (RMSE, CRPS, coverage), ablations isolating calibration from look-ahead, oracle diagnostics, a cost analysis and an honest account of the regimes where CALM does not help.

In brief, CALM reaches the final-budget integrated absolute error of SMART with 47 instead of 1000 measurements on the 3D diffusion map (21.49×), although its relative error there is worse unless $\log D$ is modelled. On the three discontinuous maps it needs 228, 247 and 241 instead of 500 (2.19, 2.02 and 2.08×). Against gSMART with a weight tuned in hindsight on the ground truth, the saving on those maps is a more modest 1.81–1.93×. Ablations show that on the discontinuous maps neither ingredient works alone: calibration without look-ahead is worse than SMART, and look-ahead without calibration is at most marginally better. On the 3D map the uncalibrated look-ahead already helps, because it avoids SMART's bias towards the boundary. As the theory predicts, CALM gives no meaningful benefit on the statistically homogeneous smooth map: its sample-efficiency ratio against SMART is 1.02, it ends level with the complete raster, and the pointwise LOO-CV criterion ends ahead.

2 Methodology

2.1 Setting and the ordinary-Kriging surrogate

An instrument measures a scalar $\rho(\mathbf{p})$ at a position $\mathbf{p}$ of a box $\mathcal{X} \subset \mathbb{R}^d$. After n measurements $\mathcal{D}_n = \{(\mathbf{p}_i, \rho_i)\}_{i=1}^n$ the experiment is summarised by an ordinary-Kriging (OK) surrogate, i.e. a Gaussian process (GP) with an unknown constant mean and the exponential covariance $C(h) = \sigma^2 e^{-h/\ell}$ (variogram $\gamma(h) = 1 - e^{-ah}$, $a = 1/\ell$, Matérn-$\frac{1}{2}$), h the Euclidean distance in raw units [Cressie, 1993, Stein, 1999, Noack et al., 2019]. We write $m_n(\mathbf{x})$ for the OK predictor, $s_n^2(\mathbf{x})$ for the OK variance and $c_n(\mathbf{x}, \mathbf{x}')$ for the OK posterior covariance. The range is refitted after every measurement with the least-squares procedure of the original SMART study: a minimises the squared misfit between $1 - e^{-ah_{ij}}$ and the squared data differences $(\rho_i - \rho_j)^2$ of all pairs, normalised by their maximum. We search $a \in [1/(10\,\mathrm{diam}),\, 1000/\mathrm{diam}]$, with diam the domain diameter. The resulting upper limit $\ell \le 10\,\mathrm{diam}$ is never reached on the 2D maps or on $\log_{10} D$, but on the two 3D maps of D it is active in 85–91% of the steps of random sampling, SMART and gSMART and in 36–44% of CALM's steps. The process variance σ^2 changes neither the predictor nor the maximiser of SMART or gSMART, and in CALM it is absorbed by the calibration mean whenever that is refitted (§2.2). It sets the scale of the predictive distribution and, in the LOO-CV criterion of §3, the relative weight of the residual term. We use its closed-form REML estimate given ℓ [Santner et al., 2003]. The reconstruction quality is measured by the integrated absolute error $\mathrm{IAE}_n = \int_{\mathcal{X}} |\rho(\mathbf{u}) - m_n(\mathbf{u})|\,\mathrm{d}\mathbf{u}/|\mathcal{X}|$.

The two published acquisitions are SMART, $\alpha_{\mathrm{S}}(\mathbf{x}) = s_n^2(\mathbf{x})$, and gSMART, $\alpha_{\mathrm{gS}}(\mathbf{x}) = s_n^2(\mathbf{x})\,(1 + c\,|\nabla m_n(\mathbf{x})|^2)$ with a user-chosen weight $c \ge 0$ [Noack et al., 2019]. For a fixed range, s_n^2 depends on the measurement *locations* only, never on the measured *values*. SMART is therefore a space-filling design whose only dependence on the data is the scalar $\hat{\ell}$ (the argmax of s_n^2 does not depend on σ^2). It cannot place more measurements where the reconstruction is actually poor. gSMART injects data dependence through the surrogate gradient, but the gradient is large on steep-but-easy (e.g. linear) regions as well as at discontinuities, and its weight c has no principled scale (§2.4).

2.2 Leave-one-out calibration of the Kriging error

Error model. We keep the stationary OK surrogate as the predictor, but replace its error model by a *locally recalibrated* one:

$$\rho(\mathbf{u}) - m_n(\mathbf{u}) \,\big|\, \mathcal{D}_n \;\approx\; \mathcal{N}\big(0,\; \lambda(\mathbf{u})\, s_n^2(\mathbf{u})\big), \tag{1}$$

with a positive, slowly varying *calibration field* λ. Equation (1) is exact for a locally scaled process $\rho(\mathbf{x}) = \mu + \tau(\mathbf{x}) g(\mathbf{x})$, g stationary with the fitted covariance, when τ is constant over the neighbourhood that screens $\mathbf{u}$; then $\lambda = \tau^2/\sigma^2$. $\lambda > 1$ flags regions where the stationary variance is over-confident (sharp features, discontinuities, steep multi-scale regions), $\lambda < 1$ regions where it is pessimistic.

Observations of λ: standardised leave-one-out residuals. For GP/Kriging interpolation the leave-one-out (LOO) residual and LOO variance at every design point are available in closed form from the Cholesky factor already maintained by the surrogate [Dubrule, 1983, Sundararajan and Keerthi, 2001]: $e_i = [\mathbf{K}^{-1}(\boldsymbol{\rho} - \mathbf{m})]_i / [\mathbf{K}^{-1}]_{ii}$, $s_{-i}^2 = 1/[\mathbf{K}^{-1}]_{ii}$, where $\mathbf{K}$ is the covariance of the OK model (including the constant-mean term). Under (1), $e_i \sim \mathcal{N}(0, \lambda(\mathbf{p}_i) s_{-i}^2)$, so $z_i^2 = e_i^2/s_{-i}^2 \sim \lambda(\mathbf{p}_i)\,\chi_1^2$ and

$$v_i \;:=\; \log z_i^2 - \mathbb{E}[\log \chi_1^2] \;=\; \log \lambda(\mathbf{p}_i) + \eta_i, \qquad \mathbb{E}\eta_i = 0,\;\; \mathrm{Var}\,\eta_i = \pi^2/2, \tag{2}$$

with $\mathbb{E}[\log \chi_1^2] = -(\gamma_{\mathrm{E}} + \log 2) \approx -1.2704$. This is the classical log-squared linearisation of stochastic-volatility models [Harvey et al., 1994]: the noise variance $\pi^2/2$ and the bias are known constants, not tuning parameters. (For numerical safety z_i^2 is floored at 10^{-12}.)

Calibration GP and the single hyperparameter-selection rule. We place a GP prior on the log-calibration field, $\log \lambda \sim \mathcal{GP}(\mu_c, \tau_c^2 e^{-|\mathbf{x}-\mathbf{x}'|/\ell_c})$, treat η_i as Gaussian with the known variance $\pi^2/2$, and fit $\boldsymbol{\theta}_c = (\tau_c^2, \ell_c, \mu_c)$ by maximising the log marginal likelihood of $(\mathbf{p}_i, v_i)_{i=1}^n$. This is the only hyperparameter-selection procedure of the method, and it is applied identically to every problem: L-BFGS-B started from the previous optimum and from two random points, within the box

$\tau_c^2 \in [10^{-4}, 10^2]$, $\ell_c \in [\mathrm{diam}(\mathcal{X})/200,\ 5\,\mathrm{diam}(\mathcal{X})]$, $\mu_c \in [-30, 30]$. The fit is repeated whenever the data set has grown by 10% since the last fit (and at $n = 10$); in between, the GP is re-conditioned on the refreshed residuals with the current $\boldsymbol{\theta}_c$. The fitted values differ between test problems, as they should; the rule that produces them does not. The posterior mean $\mu_\lambda(\mathbf{x})$ and variance $v_\lambda(\mathbf{x})$ of $\log\lambda$ give the calibration weight

$$w(\mathbf{x}) \;=\; \mathbb{E}\big[\sqrt{\lambda(\mathbf{x})}\big] \;=\; \exp\!\big(\mu_\lambda(\mathbf{x})/2 + v_\lambda(\mathbf{x})/8\big). \tag{3}$$

Two properties follow directly from the likelihood. If the residuals show no spatial structure at the resolution of the design, the likelihood either drives τ_c^2 towards zero or selects a length scale below the design spacing. In both cases w is (nearly) constant between measurements, so the acquisition below reduces to a purely geometric design: the method degrades gracefully to stationary behaviour instead of chasing χ_1^2 noise. Where the calibration is uncertain (far from data), v_λ is large and (3) contains the corresponding optimism. Both the surrogate and the calibration field are `gpCAM` GPs (§2.5).

2.3 CALM: calibrated absolute-error look-ahead

Under (1), the expected integrated absolute error of the current surrogate is $\mathbb{E}[\mathrm{IAE}_n] = \sqrt{2/\pi}\int \sqrt{\lambda(\mathbf{u})}\, s_n(\mathbf{u})\,\mathrm{d}\mathbf{u}/|\mathcal{X}|$. A new measurement at $\mathbf{x}$ updates the OK predictor by $c_n(\mathbf{u},\mathbf{x})\,(\rho(\mathbf{x}) - m_n(\mathbf{x}))/s_n^2(\mathbf{x})$. Whatever value $\rho(\mathbf{x})$ turns out to have, the conditional OK variance becomes $s_{n+1}^2(\mathbf{u}\,|\,\mathbf{x}) = s_n^2(\mathbf{u}) - c_n(\mathbf{u},\mathbf{x})^2/s_n^2(\mathbf{x})$. The *calibrated absolute-error look-ahead* (CALM) is the expected reduction of the integrated absolute error,

$$\begin{aligned} \alpha_{\mathrm{CALM}}(\mathbf{x}) &= \int_{\mathcal{X}} w(\mathbf{u})\,\Delta_n(\mathbf{u}\,|\,\mathbf{x})\,\mathrm{d}\mathbf{u}, \\ \Delta_n(\mathbf{u}\,|\,\mathbf{x}) &= s_n(\mathbf{u}) - \sqrt{\max\big\{s_n^2(\mathbf{u}) - c_n(\mathbf{u},\mathbf{x})^2/s_n^2(\mathbf{x}),\,0\big\}}, \end{aligned} \tag{4}$$

and the next measurement is $\mathbf{p}_{n+1} = \arg\max_{\mathbf{x}\in\mathcal{X}} \alpha_{\mathrm{CALM}}(\mathbf{x})$ (the constant factor $\sqrt{2/\pi}/|\mathcal{X}|$ is dropped). The integral is estimated on a fresh scrambled Sobol set [Owen, 1998] $\{\mathbf{u}_k\}_{k=1}^{M}$ drawn at every step, with $M = \min\{\max\{64n, 1024\}, 2^{15}\}$. Scaling M with n keeps the number of reference points inside the influence region of a candidate (whose volume shrinks like $|\mathcal{X}|/n$) approximately constant; the factor 64 and the memory cap 2^{15} are fixed constants of the method. α_{CALM} is set to 0 where $s_n^2(\mathbf{x}) \le 10^{-12}\,\mathrm{Var}_{\mathrm{prior}}$ (existing measurement locations).

Properties. (i) *One-step optimality.* If λ is known and locally constant over the screening neighbourhood of each candidate, the error at $\mathbf{u}$ after the update is Gaussian with variance $\lambda(\mathbf{u})s_{n+1}^2(\mathbf{u}|\mathbf{x})$. Since $\mathbb{E}|Z| = \sqrt{2/\pi}\,\mathrm{sd}(Z)$ for a centred Gaussian, (4) equals (up to a constant) $\mathbb{E}[\mathrm{IAE}_n - \mathrm{IAE}_{n+1}\,|\,\mathbf{x}]$. CALM is thus the myopic Bayes-optimal design for exactly the metric used to judge the reconstruction. (ii) *Graceful degradation.* With $w \equiv$ const, (4) is an L^1 analogue of the integrated-variance (ALC/IMSE) criteria [Cohn, 1996, Seo et al., 2000, Gramacy and Lee, 2009]. It is a purely geometric design that depends on the data only through $\hat{\ell}$, like SMART. (iii) *Self-limiting focus.* The gain of a candidate is the *area* over which it reduces the calibrated error, not the pointwise error, so refining an already dense region has diminishing returns (made quantitative in §2.4).

2.4 Allocation theory: what each criterion converges to

The following scaling analysis (derivations in Appendix B) predicts where calibration should, and should not, pay off. Assume locally $\rho \approx \mu + \tau(\mathbf{x})g(\mathbf{x})$ with g Matérn-ν ($\nu = \frac{1}{2}$ for our kernel), and let a design have local fill distance $h(\mathbf{x})$, i.e. density $\varrho(\mathbf{x}) \propto h(\mathbf{x})^{-d}$. Then $s_n(\mathbf{x}) \asymp h(\mathbf{x})^\nu$ and the expected local absolute error is $\asymp \tau(\mathbf{x})h(\mathbf{x})^\nu$.

- **SMART** equalises s_n, hence h: $\varrho_{\mathrm{S}} \propto 1$, whatever τ is.
- **Pointwise calibrated variance** (λs_n^2, or $s_n^2(1+z^2)$ as in Le Gratiet and Cannamela, 2015) equalises τh^ν: $\varrho \propto \tau^{d/\nu} = \tau^{2d}$.
- **CALM**: a greedy step equalises the marginal reduction of $\int \tau h^\nu$, the stationarity condition of minimising $\int \tau\varrho^{-\nu/d}$ subject to $\int \varrho = n$. Hence $\varrho_{\mathrm{CALM}} \propto \tau^{d/(d+\nu)} = \tau^{2d/(2d+1)}$, the IAE-optimal density.

Algorithm 1 One CALM acquisition step (all GP operations via `gpCAM`)

1: Refit the OK range $\hat{\ell}$ by least squares on the variogram cloud; $\hat{\sigma}^2$ by REML; update the Cholesky factor $\mathbf{L}$.
2: LOO residuals e_i, variances s^2_{-i} from $\mathbf{L}$; $v_i = \log \max(e_i^2/s^2_{-i}, 10^{-12}) + 1.2704$.
3: Calibration GP on $(\mathbf{p}_i, v_i)$ with noise $\pi^2/2$; refit $\boldsymbol{\theta}_c$ by maximum marginal likelihood if $n \geq 1.1\, n_{\text{last fit}}$.
4: Draw scrambled Sobol $\{\mathbf{u}_k\}_{k=1}^{M}$; compute $s_n(\mathbf{u}_k)$, $\mathbf{L}^{-1}\mathbf{k}(\mathbf{P}, \mathbf{u}_k)$ and $w(\mathbf{u}_k)$ by (3).
5: $\mathbf{p}_{n+1} \leftarrow$ differential-evolution maximiser of $\sum_k w(\mathbf{u}_k)\big[s_n(\mathbf{u}_k) - s_{n+1}(\mathbf{u}_k|\mathbf{x})\big]$ (Eq. 4).
6: Measure $\rho(\mathbf{p}_{n+1})$; append to $\mathcal{D}_n$.

Relative to the uniform design, the optimal allocation changes the error by the Hölder ratio $R = |\mathcal{X}|^{\nu/d} \int \tau \,/\, (\int \tau^{d/(d+\nu)})^{(d+\nu)/d} \geq 1$, with equality iff τ is constant, i.e. a sample-size saving of $R^{d/\nu}$. Calibration can therefore only help on functions whose local scale varies over the region resolvable by the design. It cannot help on statistically homogeneous functions, and the pointwise rule over-concentrates by the exponent $2d$ versus $2d/(2d+1)$. At a discontinuity of height J, the surrogate gradient grows like J/h while $s_n^2 \asymp h$. gSMART's score $s_n^2(1 + c|\nabla m_n|^2) \asymp h + cJ^2/h$ therefore *increases* under refinement once $cJ^2 \gtrsim h^2$: the criterion has no equilibrium and collapses onto the discontinuity (a runaway). For CALM the LOO residuals at a discontinuity stay $\approx J/2$ while $s^2_{-i} \asymp h$, so $\lambda \asymp J^2/h$ grows as well. But the candidate's gain is $\asymp \sqrt{\lambda}\, s_n\, h^d \asymp Jh^d$, which decays, so CALM keeps a finite share of measurements on smooth regions.

2.5 Implementation in gpCAM

Everything except the acquisition formulas uses `gpCAM` 7.3.4 (with its GP backend `fvgp` 4.7.9). The OK surrogate is a `gpCAM.GPOptimizer` with kernel $\sigma^2(e^{-h/\ell} + \kappa)$. The constant $\kappa = 10^4$ is a diffuse prior on the unknown mean that reproduces the OK equations (we verified agreement with the Lagrange-multiplier OK system of Noack et al., 2019 to a relative error below 3×10^{-6} for the predictor and 4×10^{-8} for the variance). A relative nugget $10^{-8}\sigma^2$ makes the surrogate numerically an interpolator. The paper's least-squares variogram fit is passed to `GPOptimizer.train` as a custom training callable. SMART, gSMART, the LOO-CV criterion of Le Gratiet and Cannamela [2015] and CALM are `gpCAM` acquisition callables. *All* of them are maximised by the same optimiser, `GPOptimizer.ask(method="global")`: differential evolution [Storn and Price, 1997] with `gpCAM`'s default settings (population $20d$, 20 generations, tolerance 10^{-6}). The only difference between methods is therefore the criterion. The calibration field is a second `GPOptimizer` with the exponential kernel, a constant mean hyperparameter and fixed noise variances $\pi^2/2$, trained with `gpCAM`'s marginal-likelihood optimiser. For a population of C candidates, (4) costs one triangular solve with the surrogate's Cholesky factor per step for the M reference points ($O(n^2 M)$), plus $O(n^2 C + nMC)$ per optimiser generation. Algorithm 1 summarises one acquisition step.

2.6 Relation to existing acquisition criteria

Table 1 places CALM among adaptive-sampling criteria for global surrogate accuracy [Liu et al., 2018, Fuhg et al., 2021]. Three ingredients have appeared separately. *Cross-validation errors* drive MEPE [Liu et al., 2017], SFCVT [Aute et al., 2013], CV-Voronoi [Xu et al., 2014], ACE [Li et al., 2010], ES-LOO [Mohammadi et al., 2022] and the LOO-adjusted variance of Le Gratiet and Cannamela [2015]. *Integrated variance reduction* drives ALC/IMSE designs [Cohn, 1996, Seo et al., 2000, Gramacy and Lee, 2009, Sauer et al., 2023]. *Gradient or Taylor-residual exploitation* drives gSMART, the feature-weighted objectives of Noack et al. [2020a], TEAD [Mo et al., 2017], LOLA-Voronoi [Crombecq et al., 2011] and GUESS [Lämmle et al., 2023]. The CV-based criteria are all *pointwise*: they maximise an (adjusted) error at the candidate, with ad hoc exploration–exploitation balances or space-filling constraints, and use raw or nearest-neighbour residuals. The integrated criteria obtain non-uniform designs only by changing the *surrogate* (treed or deep GPs), at the cost of a different, more expensive predictor. CALM differs in three respects: (a) it keeps the experimenter's stationary surrogate and moves all adaptivity into an explicitly *calibrated error model*, estimated by likelihood from the log-χ^2 structure of standardised LOO residuals, which shrinks to the stationary model when there is no evidence of heterogeneity; (b) it integrates that model over the domain in a one-step

Table 1: Acquisition criteria for global surrogate accuracy. "Data-adaptive": the criterion depends on measured values beyond global hyperparameters. "Integrated": the criterion scores the domain-wide error reduction of a candidate rather than a pointwise quantity.

Criterion	Error signal	Data-adaptive	Integrated	Free weights
SMART [Noack et al., 2019]	stationary Kriging variance	no	no	none
gSMART [Noack et al., 2019]	variance $\times(1 + c\|\nabla m\|^2)$	yes	no	c
Feature weights [Noack et al., 2020a]	variance $\times$ feature weight	yes	no	feature ranges
EIGF [Lam, 2008]	$(m - \rho_{\mathrm{nn}})^2 + s^2$	yes	no	none
MEPE [Liu et al., 2017]	LOO^2 (nearest) and variance	yes	no	adaptive balance
SFCVT [Aute et al., 2013]	surrogate of LOO errors	yes	no	space-filling bound
TEAD, GUESS [Mo et al., 2017, Lämmle et al., 2023]	Taylor residual + distance/var.	yes	no	weight schedule
LOO-CV variance [Le Gratiet and Cannamela, 2015]	$s^2(1 + e_i^2/s_{-i}^2)$, Voronoi	yes	no	none
ES-LOO [Mohammadi et al., 2022]	GP on expected sq. LOO error	yes	no	repulsion
ALC/IMSE [Cohn, 1996, Gramacy and Lee, 2009]	variance of (non-stat.) surrogate	via surrogate	yes	surrogate prior
CALM (ours)	LOO-calibrated var. (log-χ^2 GP)	yes	yes	none (ML)

look-ahead aligned with the absolute-error metric of interest, which removes the need for a separate exploration term; (c) the allocation analysis of §2.4 predicts in advance which response surfaces can benefit. The pointwise LOO criterion of Le Gratiet and Cannamela [2015] is the closest relative and is included as a baseline. Our own "CALM without look-ahead" ablation ($w(\mathbf{x})s_n(\mathbf{x})$, §5.1) isolates the value of the integration.

3 Experimental setup

Response surfaces. We use the five synthetic test functions of the original SMART study [Noack et al., 2019], implemented from its equations (Appendix A). *Piecewise-sinusoidal I and II* (Eqs. 12–13 there) on $[0, 50]^2$ add steps of $+140/+160$ and $+200/+360$ to the oscillatory base $8 - 2p_0 \sin\sqrt{|p_0|} - 2p_1 \sin\sqrt{|p_1|}$ for $p_0 \geq 30$ and for $|\mathbf{p}| \geq 20$, respectively (first matching case), mimicking phase boundaries. *Smooth oscillatory* (Eq. 15) on $[-10, 10]^2$ is $\sin 2p_0 + \sin 2p_1 + (p_0/5)^2 - (p_1/5)^2$. *Non-differentiable piecewise* (Eq. 17) on $[0, 50]^2$ is piecewise linear with a kink along $p_0 = 21.96$ for $p_1 \leq 25$ and jumps across $p_1 = 25$ and $p_0 = 21.96$ for $p_1 > 25$. The *3D diffusion coefficient* (Eq. 18) is the Stokes–Einstein law $D = k_B T/(6\pi\mu r)$ for $r \in [1, 100]$ nm, $T \in [0, 100]^\circ$C and glycerol mass fraction $C_m \in [0, 100]\%$. The viscosity $\mu(T, C_m)$ of the water–glycerol mixture is given by Cheng [2008], the reference cited by the original study. D, expressed in μm^2/s, spans 6.8 decades (from 1.7×10^{-4} to 966) and peaks in the corner $r = 1$ nm, $T = 100^\circ$C, $C_m = 0$. Two supplementary versions (Appendix F) model $\log_{10} D$ instead of D and replace Cheng's law by a simplified exponential viscosity model.

Protocol. Every run starts from 10 uniformly random points drawn from a seed-specific generator that is shared by all methods: within a seed, all methods start from the identical initial design (verified via SHA-1 hashes in the logs). Sequential methods then acquire one point at a time up to $N_{\max} = 500$ (2D) or 1000 (3D), refitting the variogram after every measurement. No method uses the total budget, so the state after N measurements is exactly that of a run stopped at N. We therefore read all budgets $N \in \{50, 100, 200, 500\}$ (2D) and $\{50, 100, 200, 500, 1000\}$ (3D) from the same trajectory, and record the error every 10 (2D) or 25 (3D) measurements to draw the convergence curves. Every method is run with 10 seeds on every function and budget. The reconstruction error is the integrated absolute error $\mathrm{IAE}_N = \frac{1}{N_{\mathrm{test}}} \sum_i |m_N(\mathbf{p}_i) - \rho(\mathbf{p}_i)|$ on a scrambled Sobol test set of 2^{14} (2D) or 2^{16} (3D) points. The test set is generated independently of all acquisitions, is never used for fitting or acquisition, and is identical across methods. For the 3D map we also report the mean absolute percentage error (MAPE). Predictive quality is assessed with the RMSE of m_N, the continuous ranked probability score (CRPS) of the Gaussian predictive distribution $\mathcal{N}(m_N, s_N^2)$ [Gneiting and Raftery, 2007], and the empirical coverage of central credible intervals at 12 nominal levels between 5% and 99%.

Methods compared. All methods share the `gpCAM` OK surrogate of §2.1; they differ only in how the next point is chosen. *Grid*: the 10 initial points plus the largest near-isotropic boundary-inclusive lattice that fits the budget (e.g. 22×22 for $N = 500$ in 2D, $10 \times 10 \times 9$ for $N = 1000$ in 3D); the remaining slots (at most 10% of the design) are filled at random so that every design

has exactly N points. *Random*: i.i.d. uniform points. *SMART*: maximum OK variance. *gSMART*: OK variance times $1 + c|\tilde{\nabla} m_N|^2$, where $\tilde{\nabla}$ is the gradient in normalised units (inputs scaled to the unit box, outputs standardised). The weight c has no prescribed value for these problems; the original study used $c = 20$ in its imaging experiment. We therefore ran the full sweep $c \in \{10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 20\}$ and report as "gSMART" the single best value, $c = 10^{-4}$. This value, chosen in hindsight on the test error, is the best or within noise of the best at $N_{\max}$ on every function (Appendix Table 11), which makes the final-budget comparison deliberately favourable to the baseline. At small budgets larger weights can be better, most strongly on the 3D map ($c = 20$ reaches 4.98 at $N = 50$ against 15.1 for $c = 10^{-4}$, before collapsing). *LOO-CV variance*: the pointwise criterion of Le Gratiet and Cannamela [2015], $s_N^2(\mathbf{x})(1 + z_i^2)$ for $\mathbf{x}$ in the Voronoi cell of $\mathbf{p}_i$. *CALM*: Eq. (4). SMART, gSMART, LOO-CV and CALM are all maximised by the identical gpCAM differential-evolution optimiser with identical settings. Ablations and oracle diagnostics are introduced in §5.1.

What counts as a speed-up. For a reference method B we define $n^\star$ as the smallest number of measurements at which the seed-averaged CALM curve reaches B's seed-averaged error at $N_{\max}$, interpolating linearly in log–log coordinates between recorded points. The *sample-efficiency ratio* is $N_{\max}/n^\star$. A speed-up is called *substantial* if the ratio is at least 2 *and* the CALM mean lies more than one standard deviation of B below B's mean at a majority of the official budgets. It is called *modest* if the ratio is at least 1.25 and CALM's mean is below B's at every budget. We also report $n^\star$ for each seed separately and whether the ± 1 s.d. bands are disjoint, so the reader can apply stricter conventions. All quantities are means $\pm$ sample standard deviations over the 10 paired seeds.

Compute. Runs were executed on a shared CPU node, one process per run and up to 84 concurrent runs. 2D runs and 3D baselines used one BLAS thread. On the main 3D map, CALM and its uncalibrated ablation used four, CALM without look-ahead and with $M = 16n$ two, and the squared-error variant and the two oracles six; CALM on the two supplementary 3D maps used two. Wall-clock times per acquisition step are logged for every step and summarised in §5.5; because of the concurrent load they are indicative rather than benchmark-grade. Code, per-step logs (selected points, measured values, surrogate and calibration hyperparameters, timings) and the scripts that generate every table and figure are provided with the paper.

4 Experimental results

4.1 Reproducing SMART and gSMART on gpCAM

The original study reports, qualitatively, that Kriging-guided sampling beats random and raster sampling and that both converge with budget [Noack et al., 2019]. Our gpCAM reproduction (Table 2, Fig. 1) confirms most of this picture and deviates from it in two places. *Confirmed.* SMART's error decreases from each official budget to the next on all five maps. It is below that of random sampling at every budget on the three discontinuous maps (e.g. 7.19 vs 9.45 on Piecewise II at $N = 500$) and from $N = 200$ on the smooth map (0.187 vs 0.357 at $N = 500$). At $N \leq 100$ the smooth map is under-resolved by every design and random sampling is ahead (by 8% at $N = 50$ and 2% at $N = 100$). On the 3D map, SMART beats the raster grid at every budget (3.10 vs 5.54 at $N = 1000$), also in MAPE, as in the original 3D study (which additionally ran a largest-unexplored-region design that we do not include). gSMART with the weight selected from our sweep ($c = 10^{-4}$, §3) improves on SMART on the three discontinuous maps and the 3D map, e.g. 2.06 vs 2.32 on Piecewise I, and matches it on the smooth map, as the original authors anticipated for sharp features. *Deviations.* First, gSMART is extremely sensitive to c (Appendix Table 11, Fig. 7). For $c \geq 10^{-2}$ the error stalls or grows with budget on the three discontinuous maps, and for $c = 10^{-3}$ it grows again after $N = 100$. With the original study's $c = 20$ it grows on four of the five maps (e.g. 62.8 at $N = 50$ and 107 at $N = 500$ on Piecewise II), the runaway predicted in §2.4, and on the smooth map it ends far above SMART (1.08 vs 0.187). Second, two cheap designs are stronger than the qualitative claim suggests. With the shared Kriging surrogate, a complete 22×22 raster matches or beats SMART at $N = 500$ on all 2D maps (Piecewise II: 6.69 vs 7.19), although it is worse on the discontinuous maps at $N = 50$ (Piecewise II: 31.4 vs 24.5). Under the original protocol of reconstructing grid designs by interpolation the raster is also competitive at $N = 500$: with linear interpolation on the discontinuous maps, with cubic interpolation on the smooth map (Appendix C). On the 3D map, i.i.d. random

Table 2: Integrated absolute error (mean ± s.d. over 10 paired seeds) on the 21 (function, budget) configurations. All methods share the `gpCAM` ordinary-Kriging surrogate and the same initial 10 points per seed. gSMART uses $c = 10^{-4}$, the best value of a 7-point sweep chosen in hindsight (the original study's $c = 20$ is in Appendix Table 11). "LOO-CV var." is the pointwise cross-validation criterion of Le Gratiet and Cannamela [2015]. Bold: CALM below every other method.

Function	N	Grid	Random	SMART	gSMART	LOO-CV var.	CALM
Piecewise I	50	10.08±0.45	12.76±1.56	8.340±0.656	8.203±0.724	8.379±1.211	**7.451±0.558**
	100	5.229±0.164	7.780±0.842	5.433±0.506	5.426±0.504	4.787±0.191	**4.433±0.193**
	200	3.910±0.070	5.149±0.647	3.735±0.184	3.625±0.208	2.838±0.127	**2.568±0.070**
	500	2.153±0.014	3.043±0.236	2.317±0.115	2.061±0.070	1.513±0.066	**1.219±0.027**
Piecewise II	50	31.39±2.01	32.45±3.23	24.49±1.30	24.15±1.32	24.43±1.87	**21.73±1.52**
	100	15.85±0.38	21.55±1.57	16.88±1.06	16.60±1.30	16.02±0.91	**14.12±0.49**
	200	12.33±0.16	15.40±1.28	11.87±0.52	11.34±0.48	10.25±0.58	**8.297±0.148**
	500	6.690±0.036	9.446±0.506	7.194±0.223	6.589±0.138	5.772±0.169	**4.118±0.080**
Smooth osc.	50	1.014±0.018	0.9445±0.0311	1.031±0.064	1.025±0.063	1.017±0.052	0.9699±0.0274
	100	1.044±0.014	0.8522±0.0200	0.8709±0.0212	0.8716±0.0193	0.8996±0.0531	**0.8002±0.0213**
	200	0.5657±0.0055	0.6871±0.0247	0.5193±0.0174	0.5192±0.0149	0.6637±0.0402	0.5215±0.0155
	500	0.1835±0.0007	0.3565±0.0152	0.1873±0.0039	0.1896±0.0031	0.1641±0.0030	0.1824±0.0037
Non-diff.	50	3.496±0.197	4.397±0.572	3.377±0.252	3.367±0.266	3.265±0.153	**3.124±0.200**
	100	3.277±0.109	2.941±0.193	2.366±0.144	2.324±0.131	2.193±0.129	**1.930±0.084**
	200	2.051±0.051	2.088±0.194	1.596±0.066	1.559±0.050	1.349±0.044	**1.142±0.042**
	500	0.9168±0.0060	1.247±0.064	0.9944±0.0380	0.9362±0.0356	0.7375±0.0122	**0.5569±0.0115**
3D diffusion	50	20.95±2.15	4.252±1.426	15.62±2.65	15.14±2.88	12.82±1.96	**3.003±0.517**
	100	16.29±1.02	3.359±0.887	11.27±1.57	11.21±1.40	6.393±0.619	**2.396±0.252**
	200	12.37±0.52	2.903±0.710	7.476±0.551	6.959±0.507	3.735±0.372	**1.810±0.104**
	500	7.824±0.358	2.076±0.190	4.596±0.209	3.818±0.216	2.167±0.172	**1.151±0.064**
	1000	5.544±0.177	1.648±0.134	3.100±0.128	2.585±0.145	1.416±0.056	**0.8641±0.0316**

sampling beats SMART at every budget (1.65 vs 3.10 at $N = 1000$). We trace this to SMART's geometry, analysed in §5.4. We report these deviations rather than tuning the baselines away: they are properties of the published criteria under a fair, shared surrogate, not of our implementation, which agrees with the textbook ordinary-Kriging equations to a relative error below 3×10^{-6} for the mean and 4×10^{-8} for the variance (`logs/test_kriging.txt`).

4.2 Reconstruction error and convergence

Table 2 lists the integrated absolute error of all methods on the 21 (function, budget) configurations; Fig. 1 shows the full convergence curves. We characterise each curve in turn.

Discontinuous maps (Piecewise I, II, non-differentiable). CALM's curve lies below every baseline from the first official budget onwards. The gap widens with n because CALM converges at a faster rate. At $N = 500$ its error is 47%, 43% and 44% lower than SMART's, and 41%, 38% and 41% lower than the tuned gSMART's. The advantage is smallest at $N = 50$ (11%, 11% and 7% below SMART, i.e. one to about two SMART standard deviations). At that stage only 40 points have been acquired and the calibration field has not yet resolved the discontinuities (§5.4). The strongest baseline at large budgets is the pointwise LOO-CV criterion, which also exploits cross-validation residuals. CALM is below it at every budget on all three maps (19%, 29% and 24% lower at $N = 500$).

Smooth oscillatory map. Here the Kriging-based designs end close together, while random sampling and the collapsed gSMART variants fall behind at large budgets. CALM is below SMART at $N = 50$ and $N = 100$ (by more than one SMART standard deviation at $N = 100$), essentially tied at $N = 200$ (relative difference 0%) and 3% lower at $N = 500$. At $N = 500$ the complete raster is level with CALM (0.184 vs 0.182). The LOO-CV criterion ends ahead (0.164) after being clearly worse at $N = 200$ (0.664 vs 0.522). CALM is thus below or level with SMART at every budget. Its advantage is largest in the under-resolved regime ($N \leq 100$) and is at most a few per cent at larger budgets, a sample-efficiency ratio of only 1.02 (§4.3); why is the subject of §5.4.

3D diffusion map. CALM is below every baseline at every budget, by a large margin except against random sampling at $N \leq 100$ (3.00 vs 4.25 at $N = 50$). At $N = 1000$ its error is 0.864 versus 3.10 for SMART, 2.58 for gSMART, 1.65 for random and 1.42 for LOO-CV. Its error at $N = 50$ (3.00) is

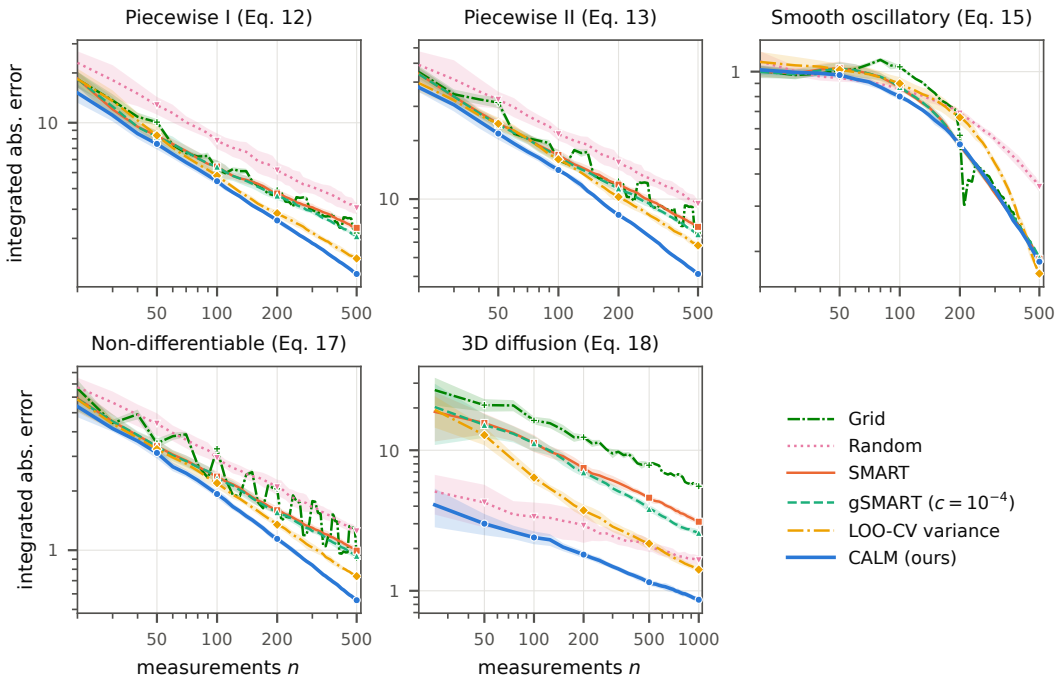


Figure 1: Convergence of the integrated absolute error (log–log) with mean ± 1 s.d. bands over 10 seeds; markers at the official budgets.

already below SMART's at $N = 1000$. Measured by the MAPE of D instead, CALM is the worst of the six main methods from $N = 200$ on, because it allocates measurements by absolute error (Table 8, §5.4). On the supplementary versions of the map (Appendix F), CALM is again the best method at every budget. On $\log_{10} D$, where the problem is far less heterogeneous, the margins are smaller: CALM reaches SMART's and gSMART's final error after 367 and 371 of 1000 measurements.

4.3 Sample efficiency

Table 3 converts the curves into the number of measurements CALM needs to reach each reference method's final-budget error. *Against SMART*, CALM is at least twice as sample-efficient on four of the five maps. It reaches SMART's final error after 228 (Piecewise I), 247 (Piecewise II) and 241 (non-differentiable) of 500 measurements, and after 47 of 1000 on the 3D map. Its mean curve is more than one SMART standard deviation below SMART's at three or more of the four or five official budgets. For the discontinuous maps the ratios (2.19, 2.02, 2.08) are close to 2. Per seed, 10, 8 and 8 of 10 seeds individually reach SMART's mean final error within 250 measurements. *Against the hindsight-tuned gSMART*, the saving on the discontinuous maps is smaller: 1.89, 1.81 and 1.93×, i.e. a modest speed-up (CALM's mean is below gSMART's at every budget, but $n^\star > N_{\max}/2$). On the 3D map it remains large (12.77×). *Against gSMART with the original weight* $c = 20$, CALM reaches the reference error within its first 50 measurements on every map. On the smooth map there is no speed-up against SMART (ratio 1.02), the tuned gSMART, the raster or LOO-CV; the only references whose final error CALM reaches early there are random sampling (1.72×, but random sampling is ahead at $N = 50$, so the class is "none") and the collapsed gSMART with $c = 20$. In summary, CALM is substantially faster than SMART on all functions with heterogeneous difficulty. It is substantially faster than every gSMART configuration on the 3D map, and 1.81–1.93× faster than the best gSMART weight found in hindsight on the discontinuous maps. It brings no meaningful benefit on the homogeneous map.

4.4 Predictive quality beyond the point estimate

Table 4 reports RMSE, CRPS and central-interval coverage at the largest budget for every method; Fig. 2 shows the full coverage curves, and Appendix Table 12 gives all budgets. RMSE follows the

Table 3: Budget CALM needs to reach a reference method's mean error at the largest budget $N_{\max}$ ($n^\star$, linear interpolation in log–log coordinates on the dense curves), sample-efficiency ratio $N_{\max}/n^\star$, the range of the per-seed ratios, the number of seeds whose own curve reaches the reference within $N_{\max}/2$, and dominance at the official budgets ($\surd$: CALM mean more than one reference s.d. below the reference mean; second column: ±1 s.d. bands disjoint). "Class" applies the definition of §3. $> N_{\max}$: the reference error is not reached within $N_{\max}$, so the ratio is below 1; $\geq$: the reference error is already reached by the 10-point initial design, so only a lower bound on the ratio is available.

Function	Reference	ref. err.	$n^\star$	ratio	per seed	seeds$\leq \frac{N_{\max}}{2}$	$>1\sigma$	disjoint	class
Piecewise I	SMART	2.32	228	2.19×	2.04–2.31	10/10	$\surd\surd\surd\surd$	$\cdot\surd\surd\surd$	substantial
	gSMART	2.06	264	1.89×	1.78–2.01	1/10	$\surd\surd\surd\surd$	$\cdot\surd\surd\surd$	modest
	gSMART, c=20	55.1	$\leq$10	$\geq$50×	$\geq$50–$\geq$50	10/10	$\surd\surd\surd\surd$	$\surd\surd\surd\surd$	substantial
Piecewise II	SMART	7.19	247	2.02×	1.95–2.08	8/10	$\surd\surd\surd\surd$	$\cdot\surd\surd\surd$	substantial
	gSMART	6.59	277	1.81×	1.75–1.89	0/10	$\surd\surd\surd\surd$	$\cdot\surd\surd\surd$	modest
	gSMART, c=20	107	$\leq$10	$\geq$50×	$\geq$50–$\geq$50	10/10	$\surd\surd\surd\surd$	$\surd\surd\surd\surd$	substantial
Smooth osc.	SMART	0.187	490	1.02×	<1–1.05	0/10	$\cdot\surd\cdot\surd$	$\cdot\surd\cdot\cdot$	none
	gSMART	0.190	485	1.03×	1.01–1.06	0/10	$\cdot\surd\cdot\surd$	$\cdot\surd\cdot\surd$	none
	gSMART, c=20	1.08	17	29.08×	23.61–36.62	10/10	$\surd\surd\surd\surd$	$\surd\surd\surd\surd$	substantial
Non-diff.	SMART	0.994	241	2.08×	1.98–2.20	8/10	$\surd\surd\surd\surd$	$\cdot\surd\surd\surd$	substantial
	gSMART	0.936	259	1.93×	1.85–2.05	3/10	$\cdot\surd\surd\surd$	$\cdot\surd\surd\surd$	modest
	gSMART, c=20	14.6	$\leq$10	$\geq$50×	$\geq$50–$\geq$50	10/10	$\surd\surd\surd\surd$	$\surd\surd\surd\surd$	substantial
3D diffusion	SMART	3.10	47	21.49×	14.40–34.26	10/10	$\surd\surd\surd\surd\surd$	$\surd\surd\surd\surd\surd$	substantial
	gSMART	2.58	78	12.77×	7.95–19.02	10/10	$\surd\surd\surd\surd\surd$	$\surd\surd\surd\surd\surd$	substantial
	gSMART, c=20	68.2	$\leq$10	$\geq$100×	$\geq$100–$\geq$100	10/10	$\surd\surd\surd\surd\surd$	$\surd\surd\surd\surd\surd$	substantial

integrated absolute error. CALM has the lowest RMSE on the three discontinuous maps (e.g. 17.2 vs 26.5 for SMART on Piecewise II) and on the 3D map. CRPS and coverage tell a different story. With the shared stationary predictive distribution $\mathcal{N}(m_N, s_N^2)$, every method is badly calibrated on the discontinuous maps: the empirical coverage of the nominal 50% interval is between 0.90 and 0.99. The single process variance $\hat{\sigma}^2$ is set by the jumps and is far too large for the smooth parts, which contain most of the test points. The methods that sample the discontinuities densely (LOO-CV and CALM) inflate $\hat{\sigma}^2$ further. They therefore have the *worst* stationary CRPS despite the best RMSE, e.g. 19.2 for CALM vs 10.7 for SMART on Piecewise II. This is a limitation of the stationary error model, not of the designs. Replacing s_N^2 by the LOO-calibrated variance $\mathbb{E}[\lambda]s_N^2$ of §2.2, fitted post hoc to each method's own design, lowers the CRPS of every design on the discontinuous and 3D maps and makes CALM's the best there (§5.3). On the smooth map, all designs except random sampling have similar RMSE and CRPS and the intervals are too wide (at $N = 500$, coverage 1.00 at every nominal level of 80% and above), reflecting the conservative exponential kernel for a smooth function.

5 Analysis

5.1 Mechanism: calibration and look-ahead are each necessary

CALM has two components, the calibrated weight w and the integrated look-ahead. We isolate them with two ablations that keep everything else identical (surrogate, optimiser, reference-set rule, seeds). *CALM without calibration* sets $w \equiv 1$, i.e. an L^1 integrated-variance design. *CALM without look-ahead* maximises the pointwise calibrated error $w(\mathbf{x})s_n(\mathbf{x})$, a smoothed, likelihood-based relative of the LOO-CV criterion. We also run two oracle diagnostics that are *not* admissible methods because they use the true function. *CALM with oracle weights* replaces $w(\mathbf{u})$ by $|\rho(\mathbf{u}) - m_n(\mathbf{u})|/(\sqrt{2/\pi}\, s_n(\mathbf{u}))$. The *greedy true-error oracle* adds, at each step, the point that most reduces the true integrated error. Table 5 and Fig. 3 summarise the results.

Neither component works alone. On the two piecewise maps the uncalibrated look-ahead is indistinguishable from SMART (Piecewise II at $N = 500$: 7.22 vs 7.19), and on the non-differentiable map it is only 3–7% better, because it is still a purely geometric design. The pointwise calibrated criterion is *worse* than SMART and has a large seed-to-seed spread (9.71±1.6). This is the over-concentration predicted by §2.4: equalising $\sqrt{\lambda}\, s_n$ drives the density towards τ^{2d} and starves the smooth regions. The combination reduces the error by 43% relative to SMART. The calibration

Table 4: Predictive metrics at $N_{\max}$ with the shared stationary predictive distribution $\mathcal{N}(m_N, s_N^2)$ (mean over 10 seeds; s.d. for MAE, RMSE and CRPS). cov_q: empirical coverage of the central q credible interval. The MAPE of the 3D map is given in Table 8.

Function	Method	MAE	RMSE	CRPS	cov_{50}	cov_{90}	cov_{95}
Piecewise I (N=500)	Grid	2.153±0.014	9.898±0.101	3.819±0.032	0.95	0.97	0.98
	Random	3.043±0.236	11.72±0.74	4.494±0.211	0.93	0.97	0.97
	SMART	2.317±0.115	10.55±0.78	3.936±0.092	0.95	0.97	0.97
	gSMART	2.061±0.070	9.273±0.455	4.277±0.086	0.96	0.98	0.98
	LOO-CV var.	1.513±0.066	6.328±0.470	11.76±0.56	0.99	1.00	1.00
	CALM	1.219±0.027	5.930±0.219	8.930±0.262	0.99	0.99	0.99
Piecewise II (N=500)	Grid	6.690±0.036	24.62±0.21	10.08±0.07	0.92	0.95	0.96
	Random	9.446±0.506	29.95±1.52	12.20±0.39	0.90	0.94	0.95
	SMART	7.194±0.223	26.52±0.94	10.74±0.28	0.92	0.95	0.96
	gSMART	6.589±0.138	23.63±0.57	11.78±0.26	0.93	0.97	0.97
	LOO-CV var.	5.772±0.169	19.80±0.64	26.29±1.59	0.98	0.99	1.00
	CALM	4.118±0.080	17.20±0.65	19.24±0.51	0.97	0.99	0.99
Smooth osc. (N=500)	Grid	0.1835±0.0007	0.2317±0.0007	0.1719±0.0002	0.93	1.00	1.00
	Random	0.3565±0.0152	0.5160±0.0229	0.2651±0.0087	0.67	0.96	0.98
	SMART	0.1873±0.0039	0.2426±0.0053	0.1741±0.0011	0.91	1.00	1.00
	gSMART	0.1896±0.0031	0.2468±0.0046	0.1750±0.0011	0.91	1.00	1.00
	LOO-CV var.	0.1641±0.0030	0.2239±0.0106	0.1704±0.0016	0.95	1.00	1.00
	CALM	0.1824±0.0037	0.2403±0.0052	0.1708±0.0011	0.92	1.00	1.00
Non-diff. (N=500)	Grid	0.9168±0.0060	3.435±0.028	1.347±0.012	0.93	0.95	0.96
	Random	1.247±0.064	4.102±0.296	1.675±0.091	0.91	0.95	0.96
	SMART	0.9944±0.0380	3.759±0.176	1.501±0.037	0.93	0.95	0.96
	gSMART	0.9362±0.0356	3.525±0.169	1.549±0.032	0.93	0.96	0.97
	LOO-CV var.	0.7375±0.0122	2.670±0.041	3.627±0.182	0.98	0.99	1.00
	CALM	0.5569±0.0115	2.377±0.073	2.752±0.076	0.98	0.99	0.99
3D diffusion (N=1000)	Grid	5.544±0.177	25.52±0.93	9.048±0.128	0.93	0.96	0.97
	Random	1.648±0.134	11.47±1.15	3.276±0.615	0.96	0.98	0.98
	SMART	3.100±0.128	14.73±1.02	6.269±0.392	0.95	0.97	0.98
	gSMART	2.585±0.145	11.06±0.59	7.993±1.081	0.96	0.99	0.99
	LOO-CV var.	1.416±0.056	5.469±0.202	13.45±0.81	0.99	1.00	1.00
	CALM	0.8641±0.0316	4.430±0.166	12.84±0.72	0.99	1.00	1.00

supplies *where* the surrogate is wrong; the integration converts it into *how many* measurements a region deserves. On the 3D map the uncalibrated look-ahead is already far better than SMART (1.38 vs 3.10 at $N = 1000$). The integrated criterion stops spending points on the faces and corners of the box (§5.4), and calibration then reduces the error by a further factor of 1.6.

Allocation and where the error goes. Fig. 4(a) tracks the fraction of measurements within 1.5 units of a discontinuity. SMART and gSMART stay near the area fraction of that band (10% on Piecewise II at $N = 500$). LOO-CV reaches 26% by $N = 100$, whereas CALM rises gradually from 18% at $N = 100$ to 42% at $N = 500$, as its calibration sharpens. Fig. 4(b) decomposes the final error of Piecewise II by distance to the discontinuity. CALM strongly reduces the error in the bands near the jump: 0.85 vs 2.67 (SMART) for test points 0.5–1.5 units away, and 0.52 vs 1.16 for 1.5–4 units. It pays for this with a larger error far from the jump (0.50 vs 0.28), exactly the trade-off the look-ahead is designed to balance. The remaining error is dominated by the innermost band (2.26), where the stationary interpolant smears any jump over the local spacing. Fig. 5 shows the calibrated error prediction itself. Where the stationary Kriging std is essentially uniform and uninformative, the calibrated prediction outlines the arc discontinuity. Across 20 quantile bins of predicted error (Fig. 4c), the calibrated prediction is monotone in the true error. Over the CALM designs of all 10 seeds at $n = 200$, its rank correlation with the true error is 0.79±0.03, against −0.38±0.03 for the stationary std. The latter is negative because on an adaptive design the stationary std is smallest where the measurements cluster, at the discontinuities, which is exactly where the error is largest.

How close is CALM to what a myopic design can do? With the true error as weights, the look-ahead improves on CALM mainly at small budgets (Piecewise II at $N = 50$: 18.8 vs 21.7). At $N = 500$ the gap is 6–7% on the discontinuous maps (e.g. 3.86 vs 4.12; Table 5). The greedy true-error oracle knows the function and maximises the actual one-step error reduction. At $N = 500$ it is at most 9% better than CALM (Piecewise I: 1.11 vs 1.22) and worse on Piecewise II (4.20). Better calibration therefore cannot buy much more for a myopic design on these maps with this surrogate: at $N = 500$, CALM is within 9% of both oracles on the discontinuous maps. On the 3D

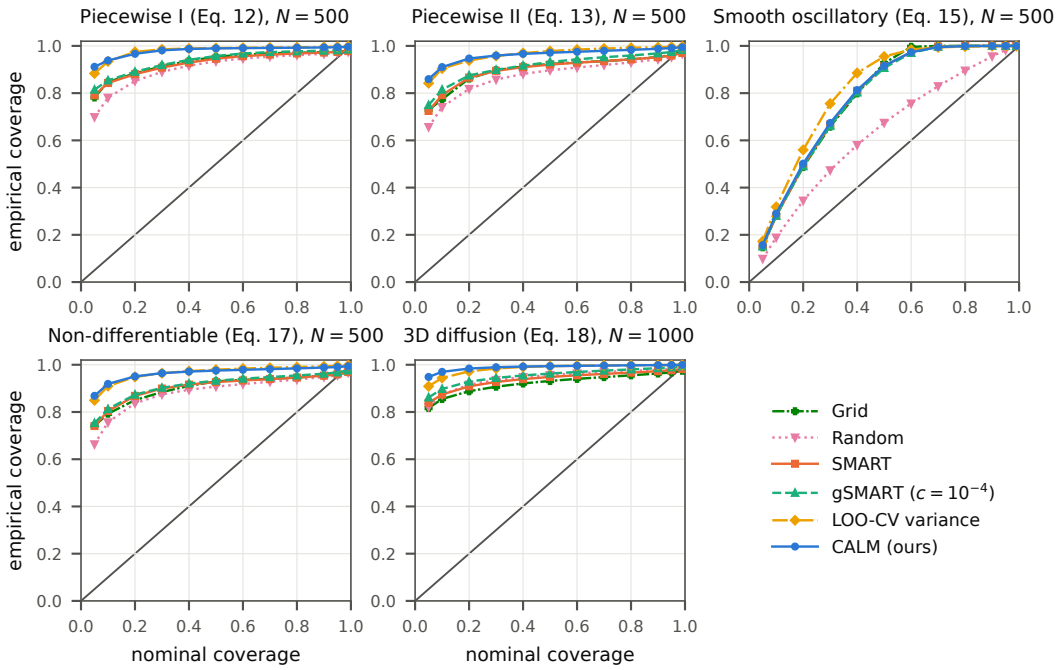


Figure 2: Coverage (reliability) curves at $N_{\max}$: empirical vs nominal coverage of central credible intervals of the stationary Kriging predictive distribution, averaged over 10 seeds. The diagonal is perfect calibration.

Table 5: Ablations and diagnostics: integrated absolute error (mean ± s.d., 10 seeds). Columns: SMART; CALM without calibration ($w \equiv 1$); CALM without look-ahead (pointwise $w\, s_n$); CALM with $M = 16n$ reference points; CALM with a squared-error look-ahead; CALM; and two oracle diagnostics that use the true function and are not admissible methods (look-ahead with oracle weights; greedy one-step reduction of the true error). The oracles indicate what better calibration or exact myopic optimisation could achieve.

Function	N	SMART	w/o calib.	w/o look-ahead	$M{=}16n$	sq.-error	CALM	oracle weights	greedy oracle
Piecewise I	100	5.433±0.506	5.562±0.221	6.706±1.201	4.441±0.155	4.350±0.139	4.433±0.193	3.752±0.162	3.640±0.138
	200	3.735±0.184	3.770±0.204	4.388±0.968	2.630±0.070	2.595±0.069	2.568±0.070	2.238±0.043	2.354±0.101
	500	2.317±0.115	2.283±0.092	3.072±0.563	1.268±0.035	1.285±0.031	1.219±0.027	1.132±0.021	1.112±0.025
Piecewise II	100	16.88±1.06	16.58±0.65	20.22±3.45	14.05±0.61	13.93±0.54	14.12±0.49	12.29±0.36	13.16±0.45
	200	11.87±0.52	11.86±0.43	16.39±3.37	8.459±0.161	8.348±0.146	8.297±0.148	7.468±0.132	8.659±0.398
	500	7.194±0.223	7.219±0.196	9.711±1.587	4.208±0.076	4.145±0.090	4.118±0.080	3.858±0.071	4.196±0.218
Smooth osc.	100	0.8709±0.0212	0.8023±0.0137	0.8889±0.0228	0.8088±0.0184	0.7998±0.0216	0.8002±0.0213	0.4440±0.0112	0.6849±0.0130
	200	0.5193±0.0174	0.5126±0.0136	0.5467±0.0431	0.5269±0.0140	0.5249±0.0135	0.5215±0.0155	0.2503±0.0026	0.5413±0.0174
	500	0.1873±0.0039	0.1820±0.0025	0.1876±0.0045	0.1847±0.0026	0.1815±0.0022	0.1824±0.0037	0.09868±0.00128	0.1776±0.0450
Non-diff.	100	2.366±0.144	2.210±0.101	2.956±0.576	1.977±0.099	1.868±0.065	1.930±0.084	1.651±0.035	1.715±0.095
	200	1.596±0.066	1.551±0.049	2.065±0.206	1.192±0.037	1.153±0.024	1.142±0.042	1.023±0.025	1.087±0.042
	500	0.9944±0.0380	0.9642±0.0289	1.261±0.198	0.5834±0.0169	0.5797±0.0149	0.5569±0.0115	0.5235±0.0088	0.5262±0.0137
3D diffusion	100	11.27±1.57	2.600±0.102	19.97±3.82	2.784±0.472	2.711±0.460	2.396±0.252	3.182±0.204	1.813±0.102
	200	7.476±0.551	2.205±0.066	14.96±2.09	2.083±0.219	2.206±0.192	1.810±0.104	2.152±0.082	1.307±0.029
	500	4.596±0.209	1.677±0.025	10.47±1.37	1.304±0.067	1.512±0.077	1.151±0.064	1.258±0.047	0.8343±0.0073
	1000	3.100±0.128	1.377±0.034	8.387±0.891	0.8727±0.0237	1.137±0.058	0.8641±0.0316	0.8289±0.0123	0.5778±0.0061

map, oracle weights do not help: they are worse than CALM up to $N = 500$ (1.26 vs 1.15) and only 4% better at $N = 1000$. The greedy true-error oracle, however, is 20–33% better than CALM at every 3D budget (0.578 vs 0.864 at $N = 1000$). It also knows the value it is about to measure, so on this map the gap comes from the unknown outcome of a measurement rather than from the error estimate; better calibration cannot close it, a surrogate that predicts D better might. The smooth map is the exception to the calibration result: there the oracle weights reduce the error by 46% at $N = 500$, because its error has structure below the scale that the residuals resolve (§5.4). Elsewhere, the remaining gap to an ideal reconstruction lies in the stationary interpolant and in what a one-step criterion can know, not in CALM's error estimate. Consistently, optimising CALM's criterion harder did not help in development. The default optimiser attains 83–91% of the best acquisition value

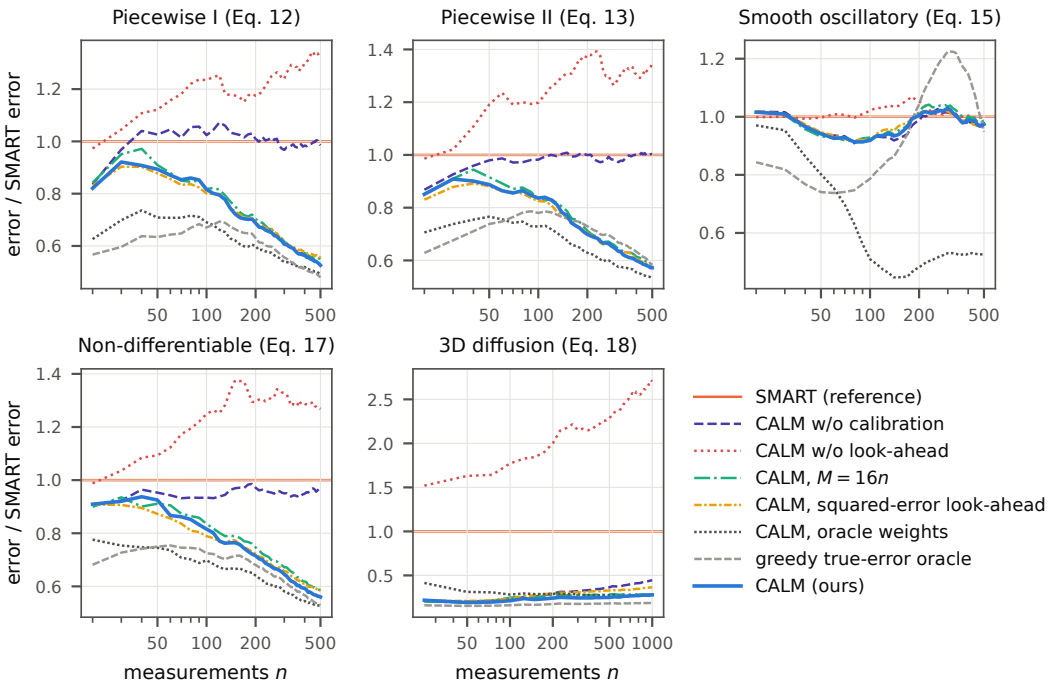


Figure 3: Error of ablations and diagnostics relative to SMART (mean over 10 seeds; below 1 is better).

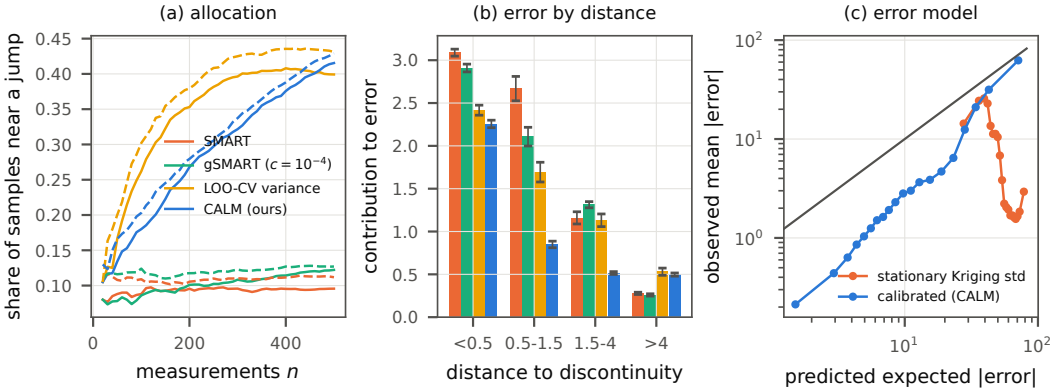


Figure 4: Mechanism. (a) Fraction of measurements within 1.5 units of a discontinuity (mean over 10 seeds; solid: Piecewise II, dashed: non-differentiable map). (b) Contribution of test points at increasing distance from the discontinuity to the integrated error at $N = 500$ (Piecewise II, mean $\pm$ s.d.; colours as in (a)). (c) Observed mean absolute error against predicted expected absolute error in 20 quantile bins, for the stationary Kriging std and for the LOO-calibrated model (Piecewise II, CALM design, $n = 200$, seed 0).

found by a dense search, yet warm-starting it from the most promising reference points did not lower the error (`attempts_log.md`, entry 2.12).

5.2 Predicting the benefit before it happens

The allocation analysis predicts the asymptotic saving from the variability of the local scale alone, $\hat{S} = \hat{R}^{2d}$ with $\hat{R}$ the power-mean ratio of the learned weights (Appendix B). This needs no ground truth: an experimenter can compute it from a space-filling pilot. Table 6 reports $\hat{S}$ computed from the calibration field fitted on SMART's final design for every seed, next to the observed sample-efficiency ratio of CALM over SMART. On the smooth map the learned field is nearly flat (largest-to-mean

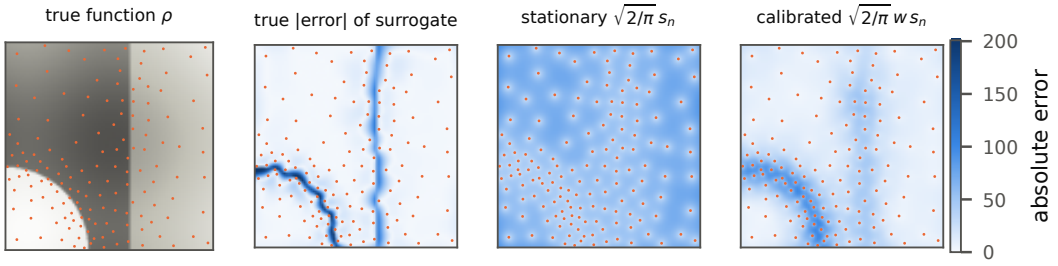


Figure 5: Piecewise II after 150 CALM measurements (seed 0): true function, true absolute error of the Kriging surrogate, the stationary error prediction $\sqrt{2/\pi}\, s_n$ and the calibrated prediction $\sqrt{2/\pi}\, w\, s_n$ used by CALM (same colour scale for the last three panels; dots are measurements).

Table 6: Predicted asymptotic sample-size saving $\hat{S}$ from the calibration field fitted on SMART's final designs (mean $\pm$ s.d. over 10 seeds) versus the observed sample-efficiency ratio of CALM over SMART (Table 3).

	Piecewise I	Piecewise II	Smooth osc.	Non-diff.	3D diffusion
predicted $\hat{S}$	3.32±0.16	2.90±0.05	1.00±0.00	2.96±0.04	6.45±0.45
observed $N_{\max}/n^\star$	2.19	2.02	1.02	2.08	21.49

weight ratio 1.0–1.4, against 9–64 on the other maps; $\hat{R} = 1.000$), and the index predicts no benefit ($\hat{S} = 1.00$); this is what we observe. The discontinuous maps are predicted to benefit by 3.32, 2.90 and 2.96×, and the 3D map most (6.45×). The ordering matches the observations, although the absolute values are asymptotic and are not reached at these budgets on the 2D maps. The calibration field thus doubles as a diagnostic that tells the experimenter, during the run, whether abandoning space filling is worthwhile.

5.3 Calibrated uncertainty as a by-product

The calibration field also improves the predictive distribution. Using $\mathcal{N}(m_N, \mathbb{E}[\lambda]\, s_N^2)$, refitted post hoc with CALM's procedure on each method's own final design, leaves the RMSE unchanged by construction. It lowers the CRPS of every Kriging design on the discontinuous and 3D maps (Table 7), e.g. from 10.7 to 6.32 for SMART and from 19.2 to 4.28 for CALM on Piecewise II. With calibrated uncertainty, CALM's designs have the lowest CRPS of all methods on these maps. The calibration can also fail as an uncertainty model. On the smooth map the marginal likelihood often selects a calibration length scale at its lower bound ($\mathrm{diam}/200$), e.g. in 8 of 10 seeds for the raster and 7 for SMART. For the irregular designs this is harmless: the test-set average of $\mathbb{E}[\lambda]$ stays between 0.65 and 0.86 and the CRPS still drops. For the raster it is not, because the bound lies far below the lattice spacing. The oscillation of period π is sampled by the lattice at only about three points per period, so its residual pattern has no smooth spatial structure at the lattice scale. Between lattice nodes the field is then unidentified and $\mathbb{E}[\lambda] = e^{\mu_\lambda + v_\lambda/2}$ is inflated by its posterior variance (test-set average 39). The CRPS of that design rises from 0.172 to 0.800. For acquisition this has little effect, because a uniform inflation does not change the maximiser of Eq. (4); for uncertainty reporting, a lower bound on ℓ_c tied to the design spacing would prevent it. The central intervals remain wider than nominal (seed-averaged coverage of the nominal 50% interval between 0.72 and 0.97 on the discontinuous and 3D maps): the error distribution near discontinuities is far heavier-tailed than any single Gaussian scale can represent.

5.4 Failure analysis: where CALM does not help, and why

Statistically homogeneous functions (smooth oscillatory map). CALM gives no speed-up on Eq. 15, and this is the expected outcome of its construction, for two independent reasons. *(i) There is no learnable heterogeneity.* The function is a sum of two stationary sinusoids and a quadratic of constant curvature, so its local difficulty is the same everywhere on scales larger than the oscillation period. The only spatial structure in the Kriging error is a modulation by the local curvature of

Table 7: CRPS at $N_{\max}$ with the stationary predictive variance s_N^2 and with the LOO-calibrated variance $\mathbb{E}[\lambda]s_N^2$ (mean over 10 seeds), for the design produced by each method.

Function	Grid	Random	SMART	gSMART	LOO-CV var.	CALM
Piecewise I	3.82 → 1.87	4.49 → 2.38	3.94 → 1.99	4.28 → 1.92	11.8 → 1.64	8.93 → 1.28
Piecewise II	10.1 → 5.97	12.2 → 7.47	10.7 → 6.32	11.8 → 6.35	26.3 → 5.91	19.2 → 4.28
Smooth osc.	0.172 → 0.800	0.265 → 0.262	0.174 → 0.166	0.175 → 0.158	0.170 → 0.149	0.171 → 0.156
Non-diff.	1.35 → 0.824	1.67 → 0.982	1.50 → 0.862	1.55 → 0.847	3.63 → 0.763	2.75 → 0.574
3D diffusion	9.05 → 4.12	3.28 → 1.26	6.27 → 2.18	7.99 → 1.85	13.5 → 1.08	12.8 → 0.695

Table 8: Mean absolute percentage error of D on the 3D map (as a fraction, mean $\pm$ s.d. over 10 seeds) for the surrogate of D itself. Every method exceeds 1 because D spans 6.8 decades and an interpolant of D has its largest relative errors where D is tiny; see Appendix F for the $\log_{10} D$ version.

Method	$N = 50$	$N = 100$	$N = 200$	$N = 500$	$N = 1000$
Grid	20.57±2.94	10.96±1.27	6.928±0.459	3.703±0.319	2.384±0.136
Random	10.84±10.85	7.183±5.510	5.187±2.029	3.496±0.849	2.463±0.661
SMART	12.60±4.04	6.390±1.201	3.480±0.526	1.830±0.215	1.240±0.070
gSMART	13.18±3.74	6.455±0.823	3.537±0.510	1.946±0.220	1.391±0.150
LOO-CV var.	40.69±17.33	13.42±3.71	5.900±0.915	2.952±0.394	2.043±0.190
CALM	13.49±8.07	13.27±5.22	10.57±2.06	8.794±2.473	7.927±1.028

$\sin 2p$, whose absolute value has period $\pi/2 \approx 1.57$. Even at $N = 500$, the mean spacing of a uniform design on $[-10, 10]^2$ is $20/\sqrt{500} \approx 0.89$, so the residuals sample this modulation at fewer than two points per period. The marginal likelihood finds almost no resolvable structure: at the end of CALM's runs the weight contrast $\max w/\bar{w}$ is 1.16 on average, against 14.6–22.1 on the discontinuous maps (Appendix Table 14). CALM therefore reduces to its geometric limit: its error matches the uncalibrated look-ahead (0.182 vs 0.182), and the heterogeneity index predicts no gain (Table 6). That structure does exist below the resolvable scale is shown by the oracle-weighted look-ahead, which knows the true error (Table 5). *(ii) Geometric limit.* For a smooth target, extrapolation error at the domain edge is first-order in the distance to the nearest point, whereas the Matérn-$\frac{1}{2}$ variance grows like its square root. The integrated look-ahead therefore values boundary candidates less than they deserve. At $N = 500$ only 3% of CALM's measurements lie within 1% of the edge, against 16% for SMART, whose variance maxima sit on the boundary. CALM's outermost 2% strip carries an error of 0.0212 versus 0.0133 for SMART, while its interior error is lower (0.143 vs 0.156; Fig. 6). The two effects roughly cancel, which is why the CALM and SMART curves end level rather than separating. At $N = 500$ the complete raster is level with CALM, and the LOO-CV criterion ends ahead after being clearly worse at $N \leq 200$. LOO-CV's advantage over SMART lies entirely in the interior (0.131 vs 0.156), and 67% of its advantage over CALM does (interior 0.131 vs 0.143; outer strip 0.0152 vs 0.0212). Its design is also less regular (coefficient of variation of nearest-neighbour distances 0.16 vs 0.12 and 0.11). We could not attribute it to a simple feature such as curvature: its share of points in the highest-curvature quartile of the domain (28%) is similar to SMART's (29%). We report it as an open observation.

Small budgets. At $N = 50$ (40 acquisitions) the calibration has seen too few residuals to separate heterogeneity reliably from χ_1^2 noise. The marginal likelihood keeps the weights weak: the weight contrast at $n = 50$ is 3.8, 2.9 and 3.0 on the three discontinuous maps, against 14.6–22.1 at $N = 500$, so CALM is still close to its geometric limit. At $N = 50$ its advantage over SMART is only 7–11% on the discontinuous maps, against more than 40% at $N = 500$. The oracle-weighted look-ahead is clearly better at that budget, so the early-stage loss is the price of statistical caution, not of the look-ahead.

Relative error in the 3D map. CALM minimises the integrated *absolute* error, and on the 3D map it spends its measurements where D is large and varies fast (small radius, hot, water-rich). The relative error where D is tiny suffers. CALM's MAPE is 7.93 at $N = 1000$, against 1.24 for SMART and 2.46 for random sampling, even though its absolute error is 21.49× more sample-efficient. This is not a defect of the look-ahead but a mismatch of objectives: if relative accuracy matters, the measured quantity should be $\log D$. On the $\log_{10} D$ version of the map (Appendix F), CALM is again the best method in both the integrated (log) error and the MAPE of D.

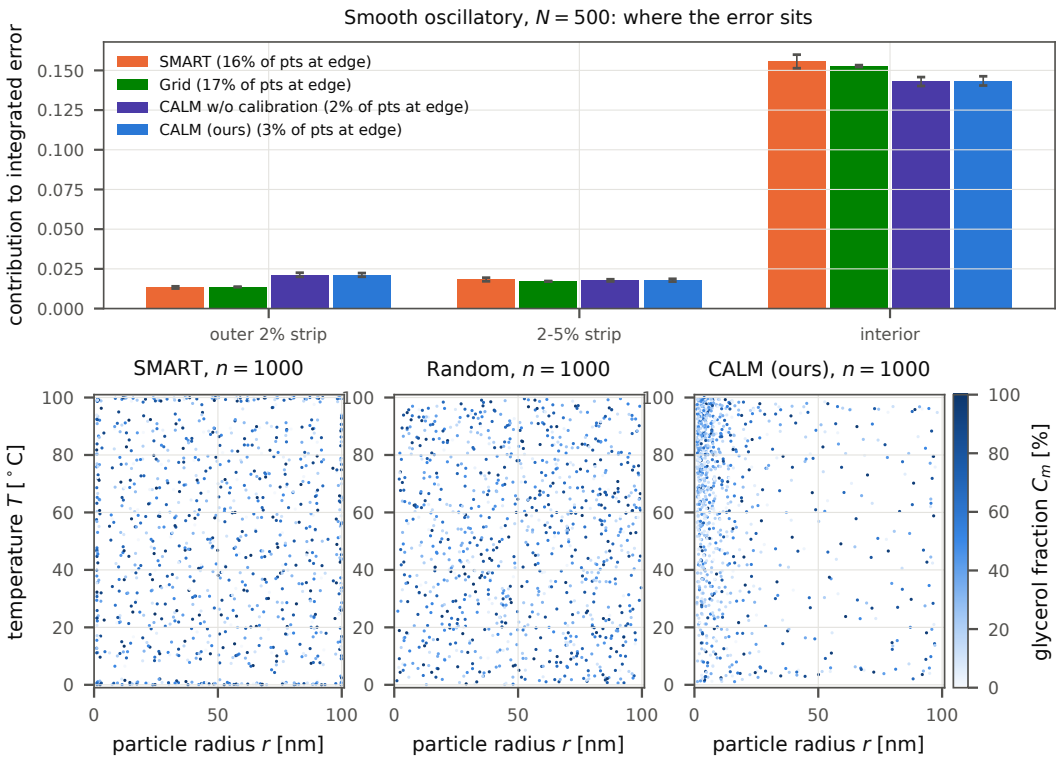


Figure 6: Top: smooth oscillatory map at $N = 500$, contribution of the outer 2% strip, the 2–5% strip and the interior to the integrated error (mean $\pm$ s.d. over 10 seeds), with the share of measurements within 1% of the edge. Bottom: 3D map, (r, T) projection of the 1000 measurements of SMART, random sampling and CALM (seed 0), coloured by glycerol fraction.

Why SMART loses to random sampling in 3D. The baseline deviation of §4.1 has the same geometric root as the smooth-map edge effect, with the opposite sign. The least-squares variogram fit returns a range far larger than the domain, so the OK variance is maximal on the boundary. In three dimensions SMART places a large share of its measurements on faces, edges and corners of the box (Fig. 6, bottom). One of those corners is the global maximum of D, which the near-linear interpolant then spreads over a wide neighbourhood. Uniform random points avoid both effects. The uncalibrated look-ahead, which scores interior coverage, already fixes most of this (§5.1).

Pathologies of the competing data-adaptive criteria. gSMART's weight has a working window of roughly one decade ($c \approx 10^{-5}$–10^{-4} in normalised units) on these maps. Larger weights collapse onto discontinuities and, on the 3D map, onto the small-radius face where $D \propto 1/r$ is steepest (97% of the measurements at $r < 5$ nm for $c = 20$; Fig. 7). Smaller weights reduce to SMART, and the useful window cannot be located without ground truth. The pointwise LOO-CV criterion follows single-cell residuals. It concentrates on discontinuities fastest (Fig. 4a), and on the smooth map it is worse than SMART at $N = 200$ (0.664 vs 0.519), consistent with χ^2_1 noise in individual residuals perturbing the design. CALM avoids both failure modes by construction, at the price of being only modestly better than the hindsight-tuned gSMART on the discontinuous maps.

5.5 Computational cost and the cost–quality trade-off

Table 9 and Fig. 8 report the wall-clock time per acquisition step, which is dominated by the optimiser's criterion evaluations (identical counts for all criteria: $20d$ individuals for ≤ 21 generations). SMART and gSMART cost 0.021 and 0.030 s per step on average on Piecewise II. CALM costs 1.7 s on average and 4.4 s over the last 50 steps of a 500-point run, i.e. 78–81 times more computation per step than SMART on the 2D maps. On the 3D map the corresponding figures are 4.2 and 4.7 s with four BLAS threads, against 0.076 s for single-threaded SMART. Two terms dominate: the

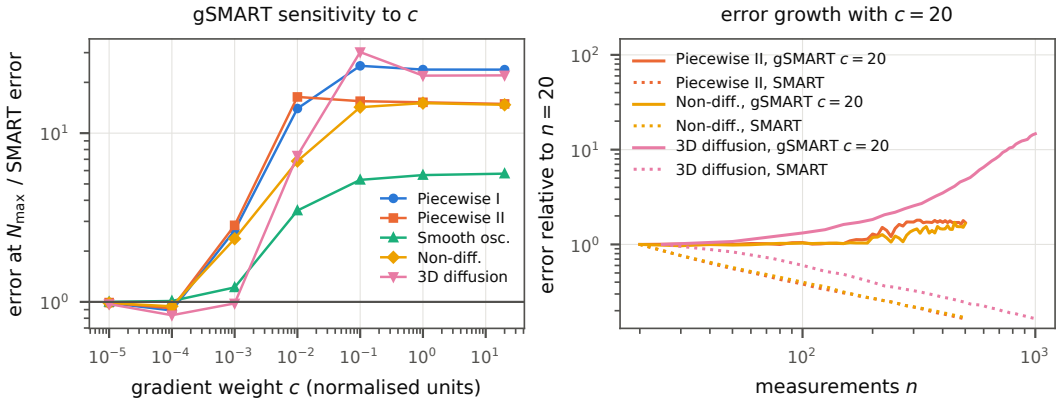


Figure 7: gSMART. Left: error at $N_{\max}$ relative to SMART as a function of the gradient weight c (normalised units; 10 seeds). Right: with the original study's $c = 20$ the error increases with the number of measurements on the discontinuous maps and on the 3D map (dotted: SMART).

Table 9: Wall-clock seconds per acquisition step (mean over all steps / mean over the last 50 steps), 10 seeds, measured under concurrent load. One BLAS thread except for the 3D runs of CALM and CALM without calibration (four threads) and of CALM without look-ahead and with $M{=}16n$ (two threads).

Function	Random	SMART	gSMART	LOO-CV var.	CALM w/o look-ahead	CALM w/o calib.	CALM $M{=}16n$	CALM
Piecewise I	0.000 / 0.00	0.021 / 0.04	0.030 / 0.05	0.024 / 0.04	0.058 / 0.11	1.109 / 2.69	0.473 / 1.28	1.685 / 4.38
Piecewise II	0.000 / 0.00	0.021 / 0.04	0.030 / 0.05	0.024 / 0.04	0.053 / 0.09	1.040 / 2.60	0.460 / 1.26	1.663 / 4.43
Smooth osc.	0.000 / 0.00	0.021 / 0.04	0.030 / 0.05	0.024 / 0.04	0.048 / 0.08	1.091 / 2.67	0.464 / 1.25	1.628 / 4.24
Non-diff.	0.000 / 0.00	0.021 / 0.04	0.030 / 0.05	0.024 / 0.04	0.055 / 0.10	1.107 / 2.73	0.477 / 1.24	1.658 / 4.41
3D diffusion	0.000 / 0.00	0.076 / 0.18	0.109 / 0.21	0.088 / 0.21	0.143 / 0.39	3.447 / 4.98	2.028 / 6.03	4.151 / 4.74

triangular solve for the M reference points, $O(n^2M)$ per step, and the $O(nMC)$ cross-covariances per generation. The reference-set size is the main cost knob. With $M = 16n$ instead of $64n$, CALM is 3.5–3.6 times cheaper on the 2D maps (Piecewise II: 0.46 vs 1.7 s per step), and its error at $N = 500$ is only 1–5% higher (Table 5, "$M{=}16n$"; Piecewise II: 4.21 vs 4.12). In absolute terms, CALM's step times (at most 7.9 s on the 2D maps and 18 s on the 3D map) are comparable to or below the time needed to move a sample and record and reduce one diffraction pattern (seconds to minutes; Noack et al., 2019). It can be hidden entirely by computing the next point while the current measurement is being taken. A two-fold reduction of the number of measurements therefore translates into an almost two-fold reduction of beam time. A slower acquisition rule that saves no measurements would be a poor trade, but this is not the case on the maps where CALM helps. On the homogeneous map, where CALM saves nothing, its extra computation is pure overhead. The heterogeneity index of §5.2 offers a principled switch: keep the cheap space-filling rule until the calibration field reports structure.

5.6 Sensitivity to the fixed constants

CALM's fitted hyperparameters are set by marginal likelihood; its remaining constants are fixed design choices, identical for all problems. The reference-set multiplier was varied ($M = 16n$ vs $64n$) in the full 10-seed protocol (Table 5). The squared-error variant of the look-ahead (weights $\mathbb{E}[\lambda]$, variance reduction instead of standard-deviation reduction) tests the choice of the L^1 objective. At $N = 500$ it is slightly worse than CALM on the discontinuous maps (Piecewise I: 1.29 vs 1.22; non-differentiable: 0.580 vs 0.557) and equal on the smooth map. On the 3D map it is 32% worse at $N = 1000$ (1.14 vs 0.864). Both results are consistent with aligning the look-ahead with the absolute-error metric. Development runs, with fewer seeds and reported only in the attempts log, explored the calibration kernel (stationary vs. non-stationary), mean trends (local spacing, extrapolation zones), the optimiser budget and warm starts, and calibration targets other than LOO residuals (local variogram increments). None improved CALM relative to the baselines; a stronger optimiser lowered the error of CALM and SMART alike. The reference-set rule $M = \min\{\max\{64n, 1024\}, 2^{15}\}$ was adopted after development runs showed Monte-Carlo noise with smaller fixed sets. It is the one constant chosen with knowledge of benchmark performance, and it was chosen once for all functions.

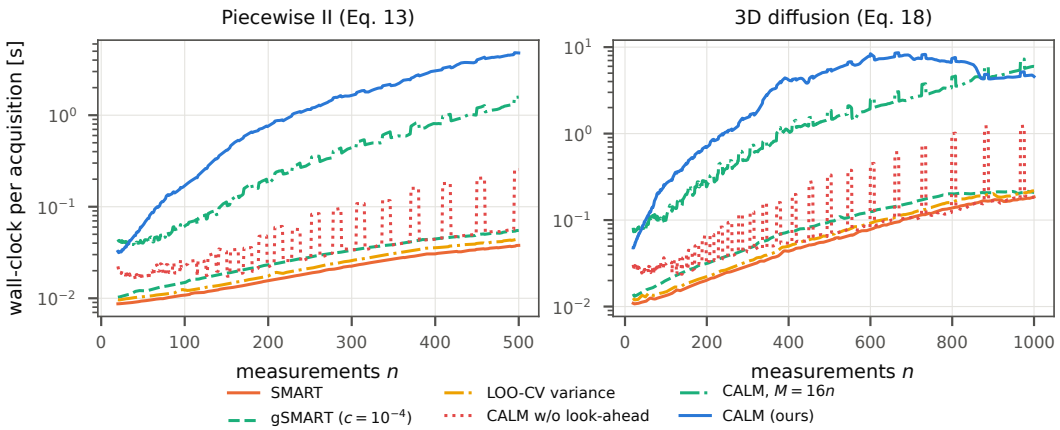


Figure 8: Wall-clock time per acquisition step against the number of measurements (10-step moving average, mean over 10 seeds).

5.7 Limitations

Our evidence is limited to the five synthetic maps of the original study, noise-free measurements and the stationary exponential-kernel Kriging surrogate. Real beamline signals are noisy. With measurement noise, the LOO residuals mix model misfit and noise, and the calibration would need a noise term. The scaling analysis is heuristic: it assumes a locally scaled Matérn process and ignores boundary effects. It predicts the ordering of the gains but not their size at finite budgets. CALM is myopic. The oracle diagnostics show that myopic design is close to saturating on our discontinuous maps, and further progress there requires non-myopic planning or a surrogate that can represent discontinuities. On the 3D map a clairvoyant greedy design is 33% better than CALM at $N = 1000$, so better one-step decisions remain possible there, although not through better calibration. Against a gSMART whose weight is tuned with ground truth, CALM's advantage on the discontinuous maps is modest (1.81–1.93×), although such tuning is unavailable in a real experiment. On homogeneous functions CALM has no advantage and costs more computation. CALM optimises absolute error and degrades relative error where the response is small, unless the response is log-transformed. Finally, the gSMART weight, the grid construction and the 3D representation required choices that the original study does not pin down; we have reported every alternative we ran.

6 Conclusion

Variance-driven Kriging acquisition, as used in SMART, decides where to measure from a quantity that does not depend on the measurements. That is why it behaves as a space-filling design and why gradient weighting, its data-dependent patch, is fragile. We kept the experimenter's Kriging surrogate and changed only the acquisition. Leave-one-out residuals, read through their log-χ^2 likelihood, calibrate the Kriging error locally. A look-ahead on the integrated absolute error then turns the calibrated error into a sampling decision. On maps with discontinuities or multi-scale structure, this halves the number of measurements needed to match SMART's integrated absolute error: four of the five benchmarks, including the physically motivated 3D diffusion map, where the saving is more than an order of magnitude. Because CALM targets absolute error, its relative error on that map is worse unless $\log D$ is modelled. Against a gSMART weight that only hindsight can find, the saving on the discontinuous maps is 1.81–1.93×. On a statistically homogeneous map it gives no meaningful benefit (sample-efficiency ratio 1.02), a regime the learned calibration field identifies during the run. Ablations and oracle diagnostics show that both ingredients are necessary on the discontinuous maps and that, at the largest budget there, CALM ends within 9% of myopic designs that know the true function. On the 3D map a clairvoyant greedy design remains 33% better at the largest budget, whereas oracle calibration weights gain only 4%. Further gains will therefore require non-myopic planning, noise-aware calibration for real beamline data, or surrogates that predict the response better (for example across discontinuities), rather than better error estimates for the same surrogate. The calibration field is cheap to compute and, wherever it is identifiable at the sampling

resolution, it also markedly improves the Kriging predictive distribution (lower CRPS for every design on the heterogeneous maps), although its central intervals remain too wide near discontinuities. We therefore suggest it as a default diagnostic for Kriging-based autonomous experiments, even where the acquisition is left unchanged.

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Scope of the appendix. The appendix contains supporting material for the main text: exact definitions, derivations, a reproduction under the original evaluation protocol, and complete versions of tables summarised in the paper. No experiment was moved here because it was incomplete. Approaches tried and abandoned during development are documented in `attempts_log.md`, not presented as results.

A Test functions

All functions are implemented in `proposal/code/testfunctions.py` and evaluated by direct calls (no lookup tables). With $b(\mathbf{p}) = 8 - 2p_0 \sin\sqrt{|p_0|} - 2p_1 \sin\sqrt{|p_1|}$: *Piecewise I* (Eq. 12) on $[0, 50]^2$ and *Piecewise II* (Eq. 13, offsets 200 and 360 instead of 140 and 160):

$$\rho = b(\mathbf{p}) + \begin{cases} 140, & p_0 \geq 30, \\ 160, & |\mathbf{p}| \geq 20 \text{ and } p_0 < 30, \\ 0, & \text{otherwise.} \end{cases}$$

Smooth oscillatory (Eq. 15) on $[-10, 10]^2$: $\rho = \sin 2p_0 + \sin 2p_1 + (p_0/5)^2 - (p_1/5)^2$. *Non-differentiable* (Eq. 17) on $[0, 50]^2$, with $p^* = 21.961143$:

$$\rho = \begin{cases} \frac{25}{12}(p_0 - 5) + 14, & p_0 > p^*,\ p_1 > 25, \\ \frac{25}{17}(p_0 - 5) + 14, & p_0 > p^*,\ p_1 \leq 25, \\ -\frac{25}{18}(p_0 - 50) + 50, & p_0 \leq p^*,\ p_1 > 25, \\ -\frac{25}{18}(p_0 - 50), & \text{otherwise.} \end{cases}$$

3D diffusion coefficient (Eq. 18):

$$D = \frac{k_B T}{6\pi\, \mu(T, C_m)\, r}, \qquad \mu = \mu_w^{\alpha} \mu_g^{1-\alpha}, \qquad \alpha = 1 - c + \frac{a\, b\, c(1-c)}{a\, c + b(1-c)}, \qquad c = C_m/100,$$

with $\mu_w = 1.790 \exp\big((-1230 - T)T/(36100 + 360T)\big)$ mPa s, $\mu_g = 12100 \exp\big((-1233 + T)T/(9900 + 70T)\big)$ mPa s, $a = 0.705 - 0.0017T$, $b = (4.9 + 0.036T)a^{2.5}$ (T in °C) [Cheng, 2008], $k_B = 1.380649 \times 10^{-23}$ J/K, T in kelvin in the numerator, $r \in [1, 100]$ nm, $T \in [0, 100]$°C, $C_m \in [0, 100]\%$; D is reported in μm^2/s. The cases of Eqs. 12–13 overlap ($p_0 \geq 30$ implies $|\mathbf{p}| \geq 20$); the first matching case is used, which reproduces the surfaces shown in Fig. 4 of the original study. The supplementary simplified viscosity is $\mu = 2.414 \times 10^{-5} \cdot 10^{247.8/(T_K - 140)}\, e^{3.5 C_m/100}$ Pa s.

B Derivations for the allocation analysis

Local error model. Let $\varrho(\mathbf{x})$ be the asymptotic design density (points per unit volume, $\int \varrho = n$) and $h(\mathbf{x}) \asymp \varrho(\mathbf{x})^{-1/d}$ the local fill distance. For a Matérn-ν covariance and $h \ll \ell$, the Kriging variance at a typical point satisfies $s_n^2 \asymp \sigma^2 (h/\ell)^{2\nu}$ (power-function bounds; Stein, 1999). Under the locally scaled model $\rho = \mu + \tau(\mathbf{x}) g(\mathbf{x})$ with slowly varying τ, the conditional error at $\mathbf{x}$ is approximately Gaussian with standard deviation $\asymp \tau(\mathbf{x}) h(\mathbf{x})^{\nu}$, so

$$\mathbb{E}[\mathrm{IAE}] \;\asymp\; \int_{\mathcal{X}} \tau(\mathbf{x})\, \varrho(\mathbf{x})^{-\nu/d}\, \mathrm{d}\mathbf{x}. \tag{5}$$

Optimal density. Minimising (5) subject to $\int \varrho = n$ gives the Lagrange condition $\frac{\nu}{d} \tau \varrho^{-\nu/d-1} = \eta$, i.e. $\varrho^\star \propto \tau^{d/(d+\nu)}$, and $\mathbb{E}[\mathrm{IAE}^\star] \asymp n^{-\nu/d} \big(\int \tau^{d/(d+\nu)} \big)^{(d+\nu)/d}$. For the uniform density $\varrho = n/|\mathcal{X}|$, $\mathbb{E}[\mathrm{IAE}_{\mathrm{unif}}] \asymp n^{-\nu/d} |\mathcal{X}|^{\nu/d} \int \tau$. Their ratio is the Hölder ratio R of §2.4, which, writing averages over $\mathcal{X}$ as $\langle \cdot \rangle$, equals $R = \langle \tau \rangle / \langle \tau^q \rangle^{1/q}$ with $q = d/(d+\nu) < 1$. This is a power-mean ratio: $R \geq 1$, with equality iff τ is constant. Because both errors scale as $n^{-\nu/d}$, matching the optimal error with a uniform design needs $R^{d/\nu}$ times more measurements.

Greedy dynamics. *SMART* adds a point where s_n is largest; at equilibrium all local maxima of s_n are equal, so h and ϱ are uniform. *Pointwise calibrated variance* adds a point where $\lambda s_n^2 \asymp \tau^2 h^{2\nu}$ is

largest; at equilibrium τh^ν is constant, so $\varrho \propto \tau^{d/\nu}$. *CALM*: adding one point to a region of volume V containing N points changes the regional contribution $\tau V (V/N)^{\nu/d}$ of (5) by $\approx \frac{\nu}{d} \tau \varrho^{-\nu/d-1}$. The greedy rule picks the largest such marginal reduction, and at equilibrium the marginal reductions are equal everywhere: $\tau \varrho^{-(d+\nu)/d} = \text{const}$, which is exactly $\varrho^\star$. CALM's one-step look-ahead is therefore a greedy algorithm for the IAE-optimal allocation. Unlike mutual information, for which greedy selection carries a near-optimality guarantee [Krause et al., 2008], the integrated L^1 criterion has no known submodularity guarantee; the oracle diagnostics of §5.1 measure the remaining gap empirically.

Discontinuities. Across a jump of height J sampled with local spacing h, the surrogate gradient is $\asymp J/h$ and $s_n^2 \asymp \sigma^2 h/\ell$ ($\nu = \frac{1}{2}$). gSMART's score $\asymp (\sigma^2/\ell)(h + c' J^2/h)$, with c' the weight in normalised units, increases as $h \to 0$ once $c' J^2 > h^2$, so new points at the jump raise its own score: there is no equilibrium. The pointwise LOO criterion $s_n^2(1 + z^2)$ has $z^2 \asymp J^2 \ell/(\sigma^2 h)$ near the jump, so its score is $\asymp J^2$, independent of h, while the score decays everywhere else. It too concentrates indefinitely, but without the gradient's self-reinforcement. CALM's local error estimate is $\sqrt{\lambda}\, s_n \asymp J$ and a candidate's gain is the integral of the error reduction over its screening region of volume $\asymp h^d$, i.e. $\asymp J h^d \to 0$. Measurements at the jump compete on equal terms with smooth regions, and the allocation stays finite.

Learned heterogeneity index. Replacing τ by the learned weight $w = \mathbb{E}[\sqrt{\lambda}]$ of Eq. (3) on a regular grid gives a data-driven estimate $\hat{R} = \langle w \rangle / \langle w^q \rangle^{1/q}$ and a predicted sample-size saving $\hat{S} = \hat{R}^{2d}$ (for $\nu = \frac{1}{2}$). These are computable during an experiment, without ground truth (§5.2).

C Reproduction under the original evaluation protocol

The original study reconstructed grid, random and largest-unexplored-region designs with piecewise-linear or cubic interpolation, and only the SMART design with Kriging. Table 10 re-evaluates our logged grid and random designs (same seeds and initial points) in that way. Linear (Delaunay) and Clough–Tocher cubic interpolation are used, with nearest-neighbour fill outside the convex hull. Clough–Tocher interpolation is defined in 2D only, so the 3D map has its own block in the table. On the 2D maps SMART beats randomly sampled designs at every budget under either reconstruction, except for a tie on the smooth map at $N = 50$ (1.03 vs 1.03). The complete 22×22 raster with linear interpolation remains competitive at $N = 500$ on the 2D maps, and with cubic interpolation it is clearly best on the smooth map. In 3D, linearly interpolated random designs beat SMART at every budget. Two differences from the original Fig. 4 limit the comparison to the ordering of SMART and random sampling. First, the errors read off that figure at $N = 500$ are about 4–13 times larger than ours, depending on the method (`logs/original_fig4_readoff.md`), although its Eq. 14 defines the same area-normalised integrated absolute error. Second, the original ranks the grid below random sampling, whereas on the same two maps (Eqs. 12 and 13) our complete rasters beat random designs at every budget under both reconstructions.

D gSMART weight sweep

Table 11 lists the error of gSMART at $N_{\max}$ for every weight of the sweep. Every weight $c \geq 10^{-2}$ is worse than SMART on every map (see also Fig. 7).

E Predictive metrics at all budgets

Table 12 extends Table 4 to every budget: RMSE, CRPS and the coverage of the central 90% interval under the stationary Kriging predictive distribution.

F Supplementary versions of the 3D map

The original study plots its 3D reconstructions on a $\log_{10} D$ colour scale, and its reported MAPE (≈ 0.1 for SMART at $N = 1000$) is only attainable if $\log D$ is modelled. Modelling D directly gives

Table 10: Integrated absolute error (mean ± s.d., 10 seeds) of grid and random designs reconstructed by interpolation, as in the original study, and of SMART with its Kriging surrogate. Top: 2D maps, linear and cubic reconstruction. Bottom: 3D map, linear reconstruction.

Function	N	Grid, linear	Grid, cubic	Random, linear	Random, cubic	SMART (Kriging)
Piecewise I	50	11.6±0.73	10.3±1.8	16.4±2.0	14.7±1.7	8.34±0.66
	100	5.48±0.13	5.85±0.46	9.83±1.2	9.60±0.61	5.43±0.51
	200	3.83±0.11	4.43±0.12	6.36±0.87	6.68±0.66	3.74±0.18
	500	2.05±0.035	2.43±0.022	3.49±0.35	3.75±0.33	2.32±0.11
Piecewise II	50	31.4±2.3	34.6±4.1	35.2±4.2	37.5±5.3	24.5±1.3
	100	14.1±0.31	18.1±1.1	23.1±2.1	25.9±1.2	16.9±1.1
	200	10.8±0.20	14.1±0.28	16.2±1.7	19.5±1.2	11.9±0.52
	500	5.73±0.074	7.69±0.064	9.50±0.67	11.4±0.66	7.19±0.22
Smooth osc.	50	0.999±0.013	1.11±0.028	1.03±0.035	1.18±0.078	1.03±0.064
	100	1.00±0.0097	1.10±0.016	0.905±0.038	0.993±0.034	0.871±0.021
	200	0.556±0.0051	0.518±0.016	0.729±0.024	0.689±0.028	0.519±0.017
	500	0.241±0.00067	0.0749±0.0012	0.418±0.017	0.235±0.016	0.187±0.0039
Non-diff.	50	2.99±0.24	3.76±0.26	4.80±0.62	5.37±0.76	3.38±0.25
	100	2.75±0.12	3.69±0.11	3.08±0.43	3.60±0.35	2.37±0.14
	200	1.73±0.053	2.35±0.094	2.09±0.36	2.55±0.20	1.60±0.066
	500	0.705±0.010	1.05±0.0070	1.17±0.11	1.53±0.092	0.994±0.038

Function	N	Grid, linear	Random, linear	SMART (Kriging)
3D diffusion	50	17.3±2.3	4.96±1.5	15.6±2.6
	100	14.4±1.0	4.30±1.2	11.3±1.6
	200	11.7±0.48	3.70±0.93	7.48±0.55
	500	7.47±0.33	2.94±0.42	4.60±0.21
	1000	5.39±0.19	2.22±0.20	3.10±0.13

Table 11: gSMART at the largest budget for every weight c of the sweep (normalised gradient units; mean ± s.d., 10 seeds). $c = 20$ is the value used by the original study in its imaging experiment.

Function ($N_{\max}$)	SMART (c=0)	c=10^{-5}	c=10^{-4}	c=10^{-3}	c=10^{-2}	c=10^{-1}	c=1	c=20
Piecewise I	2.317±0.115	2.285±0.074	2.061±0.070	6.152±0.382	32.44±22.97	58.07±9.59	55.18±12.77	55.14±12.67
Piecewise II	7.194±0.223	7.126±0.198	6.589±0.138	20.46±2.84	117.9±7.8	111.3±17.5	109.6±21.1	107.2±22.6
Smooth osc.	0.1873±0.0039	0.1872±0.0018	0.1896±0.0031	0.2281±0.0041	0.6511±0.0321	0.9898±0.1574	1.057±0.105	1.078±0.156
Non-diff.	0.9944±0.0380	0.9791±0.0296	0.9362±0.0356	2.353±0.158	6.790±1.190	14.17±4.75	14.98±4.72	14.64±5.70
3D diffusion	3.100±0.128	3.010±0.160	2.585±0.145	3.037±0.212	22.83±7.96	93.75±32.13	67.97±33.80	68.22±23.47

a MAPE above 1 for every method and budget (Table 8). We therefore repeated the 3D experiment with $\rho = \log_{10} D$ (D in m^2/s; integrated error in decades; MAPE computed on $D = 10^{\hat{\rho}}$). We also repeated it with $\rho = D$ but the simplified exponential viscosity model $\mu = \mu_w(T)e^{3.5C_m/100}$, which compresses the dynamic range to 4.5 decades. Table 13 shows that the conclusions are unchanged. CALM has the lowest mean integrated error at every budget on both versions. On $\log_{10} D$ it also has the lowest MAPE, i.e. the relative-error penalty discussed in §5.4 disappears once the measured quantity is on the scale on which accuracy is wanted. On both versions random sampling again beats SMART, confirming that this deviation is a property of SMART's geometry, not of the response scale.

G Fitted hyperparameters and their selection

All runs use the same two selection rules: the paper's least-squares variogram fit for the Kriging range (every step) and maximum marginal likelihood for the calibration field (whenever n has grown by 10%). Table 14 lists the values these rules produced at the end of the CALM runs; per-step trajectories, including the log-likelihood reached from each of the three starts, are in `logs/runs/*/calm/seed*.json` and summarised in `logs/hyperparameter_fits.json`. The smooth map is the only case where the calibration essentially switches itself off: there the fitted field has a weight contrast close to one (in 2 of the 10 seeds $\hat{\tau}_c^2$ sits at its lower bound), whereas on all other maps the contrast exceeds ten.

Table 12: RMSE / CRPS / coverage of the central 90% interval (stationary Kriging predictive distribution), mean over 10 seeds, for every (function, budget) configuration and method.

Function	N	Grid	Random	SMART	gSMART	LOO-CV var.	CALM
Piecewise I	50	24.4 / 11.4 / 0.94	25.2 / 12.2 / 0.93	20.4 / 9.85 / 0.95	20.2 / 9.85 / 0.95	16.4 / 12.4 / 0.97	17.1 / 10.1 / 0.96
	100	15.8 / 7.24 / 0.95	18.8 / 8.49 / 0.95	16.2 / 7.22 / 0.95	16.2 / 7.28 / 0.95	11.8 / 12.0 / 0.99	12.2 / 8.79 / 0.98
	200	13.9 / 5.70 / 0.96	15.4 / 6.50 / 0.96	13.5 / 5.49 / 0.96	13.1 / 5.60 / 0.96	9.31 / 12.1 / 0.99	9.15 / 8.63 / 0.99
	500	9.90 / 3.82 / 0.97	11.7 / 4.49 / 0.97	10.6 / 3.94 / 0.97	9.27 / 4.28 / 0.98	6.33 / 11.8 / 1.00	5.93 / 8.93 / 0.99
Piecewise II	50	64.5 / 31.0 / 0.89	59.5 / 29.5 / 0.92	50.6 / 24.6 / 0.93	50.0 / 24.4 / 0.93	45.3 / 29.6 / 0.97	44.1 / 24.5 / 0.95
	100	39.1 / 18.2 / 0.94	46.3 / 21.6 / 0.93	41.1 / 18.9 / 0.93	40.7 / 18.9 / 0.93	34.8 / 28.1 / 0.98	34.6 / 20.5 / 0.96
	200	35.9 / 15.4 / 0.93	39.1 / 17.2 / 0.93	34.4 / 14.8 / 0.94	32.9 / 14.9 / 0.95	27.0 / 27.1 / 0.99	25.1 / 19.2 / 0.98
	500	24.6 / 10.1 / 0.95	30.0 / 12.2 / 0.94	26.5 / 10.7 / 0.95	23.6 / 11.8 / 0.97	19.8 / 26.3 / 0.99	17.2 / 19.2 / 0.99
Smooth osc.	50	1.25 / 0.719 / 0.77	1.17 / 0.662 / 0.87	1.26 / 0.743 / 0.71	1.26 / 0.738 / 0.72	1.26 / 0.722 / 0.80	1.20 / 0.692 / 0.75
	100	1.29 / 0.793 / 0.55	1.08 / 0.598 / 0.86	1.08 / 0.613 / 0.82	1.08 / 0.613 / 0.82	1.15 / 0.635 / 0.83	1.00 / 0.559 / 0.86
	200	0.689 / 0.393 / 1.00	0.905 / 0.485 / 0.88	0.665 / 0.376 / 0.99	0.666 / 0.376 / 0.99	0.902 / 0.474 / 0.89	0.669 / 0.374 / 0.98
	500	0.232 / 0.172 / 1.00	0.516 / 0.265 / 0.96	0.243 / 0.174 / 1.00	0.247 / 0.175 / 1.00	0.224 / 0.170 / 1.00	0.240 / 0.171 / 1.00
Non-diff.	50	7.07 / 3.40 / 0.91	7.93 / 3.89 / 0.91	7.08 / 3.41 / 0.92	7.08 / 3.42 / 0.92	6.26 / 4.04 / 0.96	6.47 / 3.47 / 0.94
	100	8.62 / 3.52 / 0.90	6.38 / 2.91 / 0.92	5.94 / 2.66 / 0.93	5.81 / 2.65 / 0.93	4.83 / 3.90 / 0.98	4.80 / 2.90 / 0.96
	200	6.45 / 2.44 / 0.92	5.32 / 2.33 / 0.93	4.73 / 2.05 / 0.94	4.64 / 2.05 / 0.94	3.65 / 3.57 / 0.99	3.50 / 2.75 / 0.98
	500	3.43 / 1.35 / 0.95	4.10 / 1.67 / 0.95	3.76 / 1.50 / 0.95	3.52 / 1.55 / 0.96	2.67 / 3.63 / 0.99	2.38 / 2.75 / 0.99
3D diffusion	50	50.5 / 29.3 / 0.97	18.1 / 4.78 / 0.95	38.3 / 19.3 / 0.95	36.5 / 18.4 / 0.95	23.1 / 32.9 / 1.00	14.5 / 6.98 / 0.98
	100	48.1 / 22.5 / 0.96	16.6 / 4.58 / 0.96	32.7 / 15.1 / 0.95	32.6 / 14.8 / 0.95	14.3 / 26.3 / 1.00	12.2 / 8.16 / 0.99
	200	42.4 / 17.3 / 0.95	15.2 / 4.27 / 0.96	26.2 / 11.4 / 0.96	23.3 / 10.9 / 0.96	10.2 / 21.5 / 1.00	9.02 / 11.4 / 1.00
	500	31.4 / 12.0 / 0.96	12.6 / 3.90 / 0.98	19.2 / 7.94 / 0.97	14.7 / 8.11 / 0.98	7.39 / 16.2 / 1.00	6.34 / 11.7 / 1.00
	1000	25.5 / 9.05 / 0.96	11.5 / 3.28 / 0.98	14.7 / 6.27 / 0.97	11.1 / 7.99 / 0.99	5.47 / 13.5 / 1.00	4.43 / 12.8 / 1.00

Table 13: Supplementary 3D versions: integrated absolute error (MAPE of D in parentheses), mean over 10 seeds.

Variant	N	Grid	Random	SMART	gSMART	LOO-CV var.	CALM
3D, $\log_{10} D$	50	0.151 (0.368)	0.0975 (0.307)	0.120 (0.344)	0.119 (0.338)	0.108 (0.296)	0.0742 (0.188)
	100	0.102 (0.286)	0.0652 (0.166)	0.0847 (0.242)	0.0832 (0.234)	0.0656 (0.166)	0.0437 (0.0953)
	200	0.0725 (0.210)	0.0401 (0.0953)	0.0520 (0.144)	0.0516 (0.142)	0.0391 (0.0940)	0.0268 (0.0589)
	500	0.0399 (0.113)	0.0228 (0.0528)	0.0277 (0.0738)	0.0277 (0.0738)	0.0175 (0.0407)	0.0132 (0.0292)
	1000	0.0250 (0.0688)	0.0147 (0.0331)	0.0171 (0.0444)	0.0169 (0.0439)	0.00990 (0.0228)	0.00749 (0.0169)
3D, simple visc.	50	19.5 (4.81)	3.50 (0.867)	13.2 (2.77)	13.5 (2.84)	9.56 (4.59)	2.38 (0.719)
	100	14.2 (2.17)	2.77 (0.520)	9.31 (1.32)	9.35 (1.37)	5.39 (1.74)	1.89 (0.694)
	200	10.5 (1.17)	2.36 (0.357)	5.96 (0.659)	5.80 (0.676)	3.40 (0.817)	1.46 (0.604)
	500	6.50 (0.552)	1.69 (0.195)	3.74 (0.307)	3.19 (0.311)	1.82 (0.332)	0.964 (0.419)
	1000	4.59 (0.317)	1.35 (0.129)	2.59 (0.178)	2.19 (0.187)	1.22 (0.195)	0.691 (0.339)

H Reproducibility

All code is in `proposal/code`: `testfunctions.py` (test functions), `kriging.py` (OK on `gpCAM`), `acquisitions.py` (SMART, gSMART, LOO-CV, CALM and ablations as `gpCAM` acquisition callables), `run_experiment.py` (one run), `run_jobs.sh` (parallel launcher), `analyze.py` (tables, number macros), `figures.py`, `heterogeneity.py`, `calibrated_metrics.py`, `paper_protocol_eval.py`, `function_ranges.py` and `test_kriging.py`. Every run writes one JSON log with the initial design and its hash, every acquired point and measured value, the surrogate and calibration hyperparameters, acquisition values, optimiser evaluation counts and timings. `logs/integrity_report.txt` verifies completeness, identical initial designs across methods, and a strictly positive distance between every training point and every test point. Every experimental result in this paper is generated from these logs (`logs/numbers.tex`, `logs/tables`); values read off the original study's figures are recorded in `logs/original_fig4_readoff.md`. Development runs used to choose the method are summarised in `attempts_log.md`.

Table 14: Hyperparameters at the end of the CALM runs (mean $\pm$ s.d. over 10 seeds): Kriging range (least squares), calibration-GP variance, length scale and mean (marginal likelihood), the resulting weight contrast $\max w/\bar{w}$ on the reference set, the number of marginal-likelihood fits per run, and the median spread of the log-likelihood across the three starts of a fit.

Function	Kriging $\hat{\ell}$	$\hat{\tau}_c^2$	$\hat{\ell}_c$	$\hat{\mu}_c$	$\max w/\bar{w}$	ML fits	start spread
Piecewise I	149±2.6	7.99±0.45	7.59±0.33	−6.22±0.16	22.1±1.6	38	0.55
Piecewise II	114±2.1	10.8±0.71	10.4±0.50	−6.12±0.20	14.6±0.95	38	0.42
Smooth osc.	151±10	0.707±0.69	0.604±1.5	−0.548±0.075	1.16±0.14	38	0.0083
Non-diff.	136±3.4	10.0±0.77	9.66±0.72	−6.17±0.23	15.4±1.6	38	0.53
3D diffusion	1622±178	35.1±6.7	396±100	−1.61±0.56	79.6±15	45	0.38
3D, $\log_{10} D$	703±21	16.8±2.7	238±29	2.21±0.72	11.7±0.68	45	0.36
3D, simple visc.	1596±219	33.9±6.1	406±100	−1.51±0.50	83.2±9.6	45	0.43